On JEPA Isotropy

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Abstract

We study the empirical risk minimization (ERM) of linear Joint Embedding Predictive Architecture (JEPA) and establish its key conditions for optimality. By casting the embedding bottleneck as a rank constraint, we formulate JEPA as a reduced-rank regression problem. This induces a risk decomposition into an irreducible error and an excess-risk bound. Within the irreducible term, we find the rank of the optimal predictor saturates with the quality of the target. Within the excess-risk bound, we find spectral conditioning of the whitened end-to-end predictor governs minimization. Together, they prescribe an isotropic regularization procedure. Under this regularization, we show such JEPA isotropy subsumes canonical JEPA isotropic embedding deigns and hence justifies their empirical successes via ERM. Numerical validations corroborate our theory.

1 Introduction

We study the finite-sample risk of Joint-Embedding Predictive Architecture (JEPA) [1, 2] and establish its key conditions for optimality. JEPA has recently become a self-supervised representation learning paradigm [2–5] by jointly training a context encoder, a target encoder, and a predictor so that the predicted target embedding matches the target's.

A key recipe in JEPA training is to enforce isotropy on the encoder's marginal embedding, either through architecture [2–4] or distributional regularization [6, 5]. Their justifications for this recipe are partial: heuristic by analogy to non-predictive self-supervised learning family [7–9], passive in preventing failure modes such as representational [10, 11] or early training collapse [12], or theoretically scoped to specific probes or data assumptions [5, 13]. More fundamentally, these designs regularize only the encoder and ignore the predictor, the defining component of JEPA beyond plain Joint-Embedding Architectures (JEAs). We resolve this through a comprehensive finite-sample analysis of JEPA that jointly considers its encoder and predictor, revealing an end-to-end predictive isotropy condition that subsumes and justifies these encoder-only designs above.

Casting the embedding bottleneck as a rank constraint, we formulate multi-target JEPA as reduced-rank regression [14, 15]. This induces a decomposition of the empirical risk into an irreducible error and a finite-sample excess-risk bound (Proposition 3.16). Within the irreducible term, we identify two laws governing the rank of the optimal predictor. It grows with the number of targets K only when new targets contribute non-redundant directions, a heterogeneity condition (Theorem 3.4), and saturates at the embedding bottleneck d (Corollary 3.5 and Proposition 3.9). Any non-redundant directions beyond d contribute an irreducible error. Within the excess-risk bound, we show the spectral conditioning of the whitened end-to-end predictor T_K governs its minimization. The bound is uniquely minimized when the eigenvalues of $H_K := T_K^\top T_K$ are flat, a condition we call predictive isotropy (Lemma 3.18).

Together, the two terms prescribe an isotropic regularization on the end-to-end prediction. The irreducible term motivates target heterogeneity to fill the bottleneck; the excess-risk bound motivates flattening the spectrum of H_K. Both prescriptions reduce to regularizing T_K's spectrum toward predictive isotropy. Under this regularization, we show predictive isotropy on T_K subsumes the canonical isotropic-embedding paradigm. Through a gauge-invariant factorization of the rank-d optimum, the population-optimal encoder under this regularization automatically realizes the isotropic-Gaussian embedding that prior paradigms enforce externally (Lemma 3.21 and Theorems 3.22 and 3.23). Encoder-marginal isotropy thus emerges as a corollary of empirical risk minimization, justifying the empirical success of canonical isotropic embeddings.

Numerical experiments in a controlled linear-Gaussian setting verify each predicted phenomenon: rank growth, heterogeneity-driven recovery, the bottleneck capacity floor, and the conditioning behavior at predictive isotropy.

Contribution.

- Reduced-rank regression formulation and risk decomposition.
- Rank growth and heterogeneity in the irreducible term.
- Predictive isotropy as the excess-risk minimizer.
- A maximum-entropy regularizer on the end-to-end prediction. Isotropic-Gaussian embedding emerges as a corollary.

2 Preliminaries

In this section, we introduce the notation and background needed for our analysis. We first present the Joint-Embedding Predictive Architecture (JEPA) with frozen-teacher and linear assumptions.

Let $z \in \mathbb{R}^D$ denote a data sample drawn from a distribution $\mathcal{P}$ on $\mathbb{R}^D$. A *mask* is an index set $m \subseteq [D]$ selecting a subset of coordinates. A *mask distribution* $\mathcal{D}$ is a probability distribution over tuples $(m_c, m_{t,1}, \dots, m_{t,K})$ specifying one *context block* and $K \geq 1$ *target blocks*.[1] For a sampled tuple, write $z_{\text{ctx}} := z_{m_c} \in \mathbb{R}^{D_x}$ and $z_{\text{tgt}}^{(k)} := z_{m_{t,k}} \in \mathbb{R}^{D_y^{(k)}}$ for the corresponding context and target subvectors. JEPA jointly trains an encoder and a predictor to predict the embedding of each target block from the embedding of the context block.

Architecture. JEPA consists of three components:

(i) A *context encoder* $f_\theta : \mathbb{R}^{D_x} \to \mathbb{R}^d$ mapping the context block to an embedding $x = f_\theta(z_{\text{ctx}})$;

(ii) A *target encoder* (*teacher*) $f_{\bar{\theta}} : \mathbb{R}^{D_y^{(k)}} \to \mathbb{R}^d$ mapping each target to an embedding $y^{(k)} = f_{\bar{\theta}}(z_{\text{tgt}}^{(k)})$;

(iii) A *predictor* $g_W : \mathbb{R}^d \times \mathcal{Z} \to \mathbb{R}^d$ that predicts the k-th target embedding $y^{(k)}$ from the context embedding x and target-side metadata $q_k \in \mathcal{Z}$, i.e., $\widehat{y}^{(k)} = g_W(x, q_k)$.

Throughout this work, we assume a frozen teacher and a linear end-to-end map.

Assumption 2.1 (Frozen Teacher)**.** The target encoder $f_{\bar{\theta}}$ is fixed during training of the context encoder f_θ and predictor g_W.

Assumption 2.2 (End-to-end linearity)**.** The end-to-end map $g_W \circ f_\theta : \mathbb{R}^{D_x} \to \mathbb{R}^{Kd}$ is linear.

For explicit matrix-level analysis, we further specialize to a specific linear factorization.

Assumption 2.3 (Linear factorization)**.** The context encoder is linear, and for each target block $k \in [K]$ with fixed metadata q_k, the predictor is linear in the context embedding:

$$f_\theta(z_{\text{ctx}}) = A z_{\text{ctx}}, \quad g_W(x, q_k) = W_k x, \qquad A \in \mathbb{R}^{d \times D_x}, \quad W_k \in \mathbb{R}^{d \times d}.$$

Assumption 2.3 implies Assumption 2.2 and serves as the working factorization for Section 3.1-3.5. Section 3.6 relaxes it to nonlinear factorizations within the same end-to-end linear map (Lemma 3.21).

Under Assumption 2.3, stacking the target embeddings,

$$Y := \left[(y^{(1)})^\top \quad \cdots \quad (y^{(K)})^\top \right]^\top \in \mathbb{R}^{Kd}, \tag{1}$$

$$\widehat{Y} = \begin{bmatrix} \widehat{y}^{(1)} \\ \vdots \\ \widehat{y}^{(K)} \end{bmatrix} = Wx, \qquad W := \begin{bmatrix} W_1 \\ \vdots \\ W_K \end{bmatrix} \in \mathbb{R}^{Kd \times d}. \tag{2}$$

Thus the end-to-end map from context to predicted target embeddings is $WA \in \mathbb{R}^{Kd \times D_x}$.

Below we define the JEPA training objective. Unless otherwise stated, all expectations are taken over $z \sim \mathcal{P}$ and $(m_c, m_{t,1}, \dots, m_{t,K}) \sim \mathcal{D}$.

[1] We call a mask a *block mask* when it selects a contiguous index set. The target blocks need not be disjoint.

Definition 2.4 (JEPA loss)**.** Let $x := Az_{\text{ctx}} \in \mathbb{R}^d$ and stack the target embeddings $y^{(k)} := f_{\bar{\theta}}(z_{\text{tgt}}^{(k)})$ into Y as in (1), and the predictor blocks W_k into W as in (2). The K-target JEPA loss is

$$\mathcal{L}_K(W, A) \;:=\; \frac{1}{K}\,\mathbb{E}\big[\|Y - Wx\|^2\big]. \tag{3}$$

The composition WA acts on the raw context x only through its row space, a low-dimensional subspace of $\mathbb{R}^{D_x}$ that captures the directions JEPA effectively operates on. This subspace is central to our analysis, so we name it next.

Definition 2.5 (Predictive subspace)**.** A linear subspace $V \subseteq \mathbb{R}^{D_x}$ is called a *predictive subspace* if $V = \mathrm{row}(WA)$ for some admissible predictor WA under Assumption 2.3. We write $\mathcal{V}_r$ for the set of predictive subspaces of dimension r.

Denote the population second moments $\Sigma_{XX} := \mathbb{E}[xx^\top] \in \mathbb{R}^{d\times d}$, $\Sigma_{Y^{(k)}X} := \mathbb{E}[y^{(k)}x^\top] \in \mathbb{R}^{d\times d}$ for $k \in [K]$, and their stack $\Sigma_{YX} := \mathbb{E}[Yx^\top] = [\Sigma_{Y^{(1)}X}^\top, \dots, \Sigma_{Y^{(K)}X}^\top]^\top \in \mathbb{R}^{Kd\times d}$. Assume $\Sigma_{XX} \succ 0$ (non-degenerate context embedding).

3 Methodology

3.1 JEPA as Reduced-Rank Regression

We show that, under the frozen-teacher and linearity (Assumptions 2.1 and 2.3), JEPA reduces to a reduced-rank regression problem. For $r \le d$, consider the rank-constrained JEPA problem

$$W_r^\star \in \arg\min_{W\in\mathbb{R}^{Kd\times d}} \mathcal{L}_K(W, A) \qquad \text{subject to} \qquad \mathrm{rank}(W) \le r, \tag{4}$$

where $\mathcal{L}_K(W, A)$ is the JEPA loss in Definition 2.4.

Proposition 3.1 (JEPA as reduced-rank regression)**.** *Under Assumptions 2.1 and 2.3, the optimization problem* (4) *is exactly the reduced-rank regression problem in Definition A.1 with* $p = d$*,* $q = Kd$*, predictor* $x \in \mathbb{R}^d$*, and response* $Y \in \mathbb{R}^{Kd}$*.*

Let $B_{\text{OLS}} := \Sigma_{YX}\Sigma_{XX}^{-1}$ from (17). By Lemma A.2, the closed-form solution of (4) is $W_r^\star = B_{\text{OLS}}\Sigma_{XX}^{1/2}V_rV_r^\top\Sigma_{XX}^{-1/2}$, where $V_r \in \mathbb{R}^{d\times r}$ collect the top-r eigenvalues of $\Sigma_{XX}^{1/2}B_{\text{OLS}}^\top B_{\text{OLS}}\Sigma_{XX}^{1/2}$.

Below we study when increasing the number of target blocks can enlarge the predictive structure accessible from the context embedding x.

3.2 Multi-Target Masking and Rank Growth

The JEPA predictor in (4) is subject to two rank bottlenecks: the explicit constraint r, and an intrinsic bottleneck induced by the data distribution through x. Lemma 3.2 makes the latter precise:

Lemma 3.2 (Rank of the JEPA solution)**.** *Let $W_r^\star$ be the solution to Proposition 3.1. Then*

$$\mathrm{rank}(W_r^\star) = \min\{r,\, \mathrm{rank}(\Sigma_{YX})\}. \tag{5}$$

Proof. See Section B.1 for a detailed proof. □

Lemma 3.2 identifies $\mathrm{rank}(\Sigma_{YX})$ as the intrinsic bottleneck predictive rank: adding target blocks enlarges the predictive structure only insofar as it raises this rank. To study how $\mathrm{rank}(\Sigma_{YX})$ depends on the choice and number of target blocks, we aggregate across blocks. Recall $\Sigma_{Y^{(k)}X}, \Sigma_{YX}$ defined in Section 2. It is natural to define the predictor-side Gram matrix

$$G_K := \Sigma_{YX}^\top\Sigma_{YX} = \sum_{k=1}^{K}\Sigma_{Y^{(k)}X}^\top\Sigma_{Y^{(k)}X} \in \mathbb{R}^{d\times d}. \tag{6}$$

Each summand $\Sigma_{Y^{(k)}X}^\top\Sigma_{Y^{(k)}X}$ is positive semidefinite, so adding a block can only enlarge the range of G_K (equivalently, shrink its kernel). Throughout, we write $\lambda_i(\cdot)$ and $\sigma_i(\cdot)$ for the i-th largest eigenvalue and singular value (in decreasing order), respectively, and $\|\cdot\|_{\text{op}}$ for the operator norm. Since $G_K = \Sigma_{YX}^\top\Sigma_{YX}$, its eigenvalues coincide with squared singular values of Σ_{YX}; in particular, $\lambda_i(G_K) = \sigma_i(\Sigma_{YX})^2$ for every i.

Definition 3.3 (Predictive rank)**.** For a given number of target blocks K, the *predictive rank* is

$$r_\star(K) := \mathrm{rank}(\Sigma_{YX}).$$

Since $\Sigma_{XX} \succ 0$, $G_K = \Sigma_{YX}^\top\Sigma_{YX} \in \mathbb{R}^{d\times d}$, one also has $r_\star(K) = \mathrm{rank}(G_K) \le d$. Intuitively, $r_\star(K)$ is the predictive subspace dimension (Definition 2.5). We next study how it depends on K.

Theorem 3.4 (Rank growth). *Let $G_K := \Sigma_{YX}^\top \Sigma_{YX}$ and $r_\star(K) := \operatorname{rank}(\Sigma_{YX})$ for $K \geq 1$. Then:*

1. ***Monotonicity.***
$$r_\star(K+1) \geq r_\star(K).$$
2. ***Strict growth under complementarity.*** *If $\ker(G_K) \not\subseteq \ker(\Sigma_{Y^{(K+1)}X})$, then*
$$r_\star(K+1) > r_\star(K).$$

Proof. See Section B.2 for a detailed proof. □

Corollary 3.5 (Saturation). *There exists $K^\star$ such that $r_\star(K) = r_\star(K^\star) \leq d$ for every $K \geq K^\star$.*

Corollary 3.5 is trivial from Theorem 3.4. Besides the predictive rank Definition 3.3, it is useful to track how far G_K is from being rank-deficient on its range, i.e., its smallest nonzero eigenvalue.

Definition 3.6 (Heterogeneity level). The heterogeneity level λ_{het} is the smallest nonzero eigenvalue of G_K, equivalently $\lambda_{r^\star(K)}(G_K)$.

By the identity $G_K = \Sigma_{YX}^\top \Sigma_{YX}$, this is the squared smallest nonzero singular value of Σ_{YX}
$$\sigma_{r_\star(K)}(\Sigma_{YX}) = \sqrt{\lambda_{\mathrm{het}}}. \tag{7}$$

Remark 3.7 (Heterogeneity as complementarity). Rank increases only if the new block cross-covariance $\Sigma_{Y^{(K+1)}X}$ is nonzero on $\ker(G_K)$. Equivalently, the new block must reveal directions of X invisible to the previously accumulated structure; highly overlapping or redundant target blocks fail this condition. In this sense, heterogeneity means complementarity among blockwise cross-covariances, not merely a larger number of targets.

The next proposition shows that λ_{het} improves how well the top-$r^\star(K)$ right singular subspace of Σ_{YX} is identified from finite samples.

Proposition 3.8 (Heterogeneity controls predictive-subspace identifiability). *Assume the family $\{\Sigma_{Y^{(k)}X}\}_{k=1}^K$ has heterogeneity level $\lambda_{\mathrm{het}} > 0$ (Definition 3.6), and let $\widehat{\Sigma}_{YX}$ be the empirical cross-covariance computed from n i.i.d. samples under the sub-Gaussian assumptions of Lemma A.8. Let $V_{r_\star(K)}, \widehat{V}_{r_\star(K)}$ denote the top-$r_\star(K)$ right singular subspaces of Σ_{YX} and $\widehat{\Sigma}_{YX}$ respectively. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$,*
$$\|\sin\Theta(V_{r_\star(K)}, \widehat{V}_{r_\star(K)})\|_{\mathrm{op}} \leq \frac{\epsilon_n}{\sqrt{\lambda_{\mathrm{het}}} - \epsilon_n}, \quad \epsilon_n := C\|\Sigma_{YX}\|_{\mathrm{op}}\sqrt{\tfrac{\mathrm{effdim}(\Sigma_{YX}) + \log(1/\delta)}{n}},$$
whenever $\epsilon_n < \sqrt{\lambda_{\mathrm{het}}}$.

Proof. See Section B.3 for a detailed proof. □

For sufficiently large n such that $\epsilon_n \leq \sqrt{\lambda_{\mathrm{het}}}/2$, the bound simplifies to $\widetilde{O}\left(1/\sqrt{n\lambda_{\mathrm{het}}}\right)$, so stronger heterogeneity (larger λ_{het}) tightens the $\sin\Theta$ bound and yields better identifiability.

3.3 Capacity Limits of the Embedding Bottleneck

This subsection studies the architectural bottleneck itself. Since JEPA predicts through a d-dimensional representation $x = Az_{\mathrm{ctx}}$, its end-to-end predictor from the raw context block to the stacked target embedding must factor through a rank at most d map. This induces an approximation limit even when the raw context contains richer predictive structure. For convenience, we denote
$$Z := z_{\mathrm{ctx}} \in \mathbb{R}^{D_x}, \qquad \Sigma_{ZZ} := \mathbb{E}[ZZ^\top], \qquad \Sigma_{YZ} := \mathbb{E}[YZ^\top],$$
and assume $\Sigma_{ZZ} \succ 0$. Define the unconstrained population linear predictor from raw context to stacked targets by $B_{\mathrm{OLS}}^{\mathrm{raw}} := \Sigma_{YZ}\Sigma_{ZZ}^{-1} \in \mathbb{R}^{Kd \times D_x}$. The next result formalizes the irreducible approximation error imposed by the d-dimensional embedding bottleneck.

Proposition 3.9 (Capacity lower bound from the embedding bottleneck). *Let $T_K := B_{\mathrm{OLS}}^{\mathrm{raw}}\Sigma_{ZZ}^{1/2}$ and $\sigma_1(T_K) \geq \sigma_2(T_K) \geq \cdots \geq 0$ denote its singular values. Then for every $(W, A) \in \mathbb{R}^{Kd\times d} \times \mathbb{R}^{d \times D_x}$,*
$$\mathbb{E}\big[\|Y - WAZ\|_2^2\big] \geq \mathbb{E}\big[\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}}Z\|_2^2\big] + \sum_{j>d}\sigma_j(T_K)^2. \tag{8}$$

Proof. See Section B.4 for a detailed proof. □

Remark 3.10 (The capacity term). The bound $\mathcal{E}_{\mathrm{tail}}(d) = \sum_{j>d}\sigma_j(T_K)^2$ is architecturally irreducible: no choice of (W, A) can recover the predictive content beyond the top-d singular directions of T_K. The spectral profile $\{\sigma_j(T_K)\}_{j>d}$ thus quantifies the unavoidable cost of the embedding bottleneck.

Proposition 3.9 sharpens the saturation of Section 3.2: the embedding dimension d is an architectural ceiling that additional target blocks cannot bypass. The next subsection studies the finite-sample room for improvement within this population ceiling.

3.4 Post-Saturation Gains via Eigenvalue Balancing

By Corollary 3.5, $r_\star(K)$ eventually saturates at some value at most d. Additional targets cannot enlarge the accessible predictive subspace (Definition 2.5), but at fixed rank, finite-sample recovery still depends on how that rank is conditioned. Below we focus on the regime in which the saturated value reaches the architectural ceiling $r_\star(K) = d$, and identify the governing conditioning quantity.

Given $T_K = B_{\mathrm{OLS}}^{\mathrm{raw}} \Sigma_{ZZ}^{1/2}$ from Proposition 3.9 and let $H_K := T_K^\top T_K$, define

$$\kappa_K^2 := \frac{\lambda_d(H_K)}{\lambda_1(H_K)} = \left(\frac{\sigma_d(T_K)}{\sigma_1(T_K)}\right)^2. \tag{9}$$

where κ_K^2 is scale-invariant and satisfies $0 < \kappa_K^2 \le 1$.

Assumption 3.11 (Finite-window post-saturation conditioning)**.** There exist constants $K^\star \le K_{\max}$, $\kappa_0 > 0$, $\gamma > 0$, and $r_{\mathrm{eff}} > 0$ such that for every $K \in \{K^\star, \dots, K_{\max}\}$,

$$r_\star(K) = d, \qquad \operatorname{rank}(T_K) = d, \qquad \kappa_K^2 \ge \kappa_0^2 + \gamma(K - K^\star), \qquad \operatorname{effdim}(T_K) \le r_{\mathrm{eff}}.$$

Let $V_K, \widehat{V}_K \in \mathcal{V}_d$ denote the top-d right singular subspaces of T_K and $\widehat{T}_K$, respectively. To translate conditioning-driven subspace stability into excess risk, we assume a Lipschitz relation between the two. For a predictive subspace $V \in \mathcal{V}_d$, define

$$\mathcal{R}_K(V) := \frac{1}{K} \inf_{W,A:\, \mathrm{row}(WA) \subseteq V} \mathbb{E}\big[\|Y - WAZ\|^2\big]\,, \tag{10}$$

the optimal per-target risk among predictors using V. The factor $1/K$ removes trivial growth in target dimension, so variation in $\mathcal{R}_K$ reflects predictive quality rather than aggregation.

Assumption 3.12 (Subspace-Lipschitz excess risk)**.** Let $\mathcal{R}_K^\star := \mathcal{R}_K(V_K)$. There exists a constant $L > 0$ such that for every $K \in \{K^\star, \dots, K_{\max}\}$,

$$\mathcal{R}_K(\widehat{V}_K) - \mathcal{R}_K^\star \le L\,\|\sin\Theta(V_K, \widehat{V}_K)\|_{\mathrm{op}}. \tag{11}$$

Based on Assumptions 3.11 and 3.12, we obtain a risk decomposition of the per-target risk into a population floor and a finite-sample excess term, where the derivation deferred to Section A.3.

Proposition 3.13 (Post-saturation risk decomposition)**.** *Under Assumptions 3.11 and 3.12 and the half-gap condition (23), for any $K \in \{K^\star, \dots, K_{\max}\}$ and $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\mathcal{R}_K(\widehat{V}_K) \le \underbrace{\tfrac{1}{K}\big(\mathbb{E}\big[\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}} Z\|^2\big] + \mathcal{E}_{\mathrm{tail}}(d)\big)}_{\mathcal{R}_K^\star,\ \textit{population rank-d risk}} + \underbrace{\frac{2LC}{\sqrt{\kappa_0^2 + \gamma(K - K^\star)}}\sqrt{\frac{r_{\mathrm{eff}} + \log(1/\delta)}{n}}}_{\textit{excess risk}}. \tag{12}$$

Proof. See Section B.5 for a detailed proof. □

The excess-risk term in (12) is governed by two finite-sample effects: shrinkage in n at fixed K (Remark 3.14), and post-saturation conditioning gains in K at fixed n (Remark 3.15).

Remark 3.14 (Sample complexity). Inverting the excess risk (30), we obtain $\mathcal{R}_K(\widehat{V}_K) - \mathcal{R}_K^\star \le \varepsilon$ with probability at least $1-\delta$ requires $n = \widetilde{\mathcal{O}}\big(L^2\, r_{\mathrm{eff}} / \big(\kappa_0^2 + \gamma(K - K^\star)\big)\, \varepsilon^{-2}\big)$ samples. The rate is parametric in ε; the only lever on the prefactor is the post-saturation conditioning $\kappa_0^2 + \gamma(K - K^\star)$ of the whitened end-to-end predictor T_K, deferred to Section 3.6.

Remark 3.15 (Second saturation). Corollary 3.5 establishes a population saturation of $r_\star(K)$. Even when this saturation occurs at the architectural ceiling $r_\star = d$, Proposition 3.13 establishes a second, finite-sample saturation: recovery continues to improve as new blocks balance the eigenvalue-conditioning of H_K, until κ_K^2 stops improving, with absolute ceiling 1.

3.5 Risk Reduction in Multi-Target JEPA

The previous subsections identify two distinct mechanisms by which additional target blocks improve JEPA. Theorem 3.4 and Proposition 3.8 identify the pre-saturation lever: heterogeneity of $\{\Sigma_{Y^{(k)}X}\}_{k=1}^K$ controls both the population predictive rank and its finite-sample identifiability. Proposition 3.13 identifies the post-saturation lever: once $r_\star(K) = d$, additional targets improve recovery only by balancing the spectrum of $T_K := B_{\mathrm{OLS}}^{\mathrm{raw}} \Sigma_{ZZ}^{1/2}$. Let $V_{K,r}, \widehat{V}_{K,r} \in \mathcal{V}_r$ denote the top-r right singular subspaces of T_K and $\widehat{T}_K$, respectively. The next result unifies both into a single risk.

Proposition 3.16 (Unified finite-sample risk decomposition)**.** *Fix K and $r \le d$. Let $\Delta_{K,r} :=$ $\sigma_r(T_K) - \sigma_{r+1}(T_K)$, with the convention $\sigma_{r+1}(T_K) = 0$ when $r = \operatorname{rank}(T_K)$, and let $\epsilon_n(T_K) :=$ $C\|T_K\|_{\mathrm{op}}\sqrt{(\operatorname{effdim}(T_K) + \log(1/\delta))/n}$. Assume $\epsilon_n(T_K) < \Delta_{K,r}$, and suppose the event*

$$\mathcal{E} := \left\{ \|\widehat{T}_K - T_K\|_{\mathrm{op}} \le \epsilon_n(T_K) \right\}$$

holds with probability at least $1 - \delta$. Then, under rank-r version of Assumption 3.12, on the event $\mathcal{E}$,

$$\mathcal{R}_K(\widehat{V}_{K,r}) \le \underbrace{\frac{1}{K}\left(\mathbb{E}\big[\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}} Z\|^2\big] + \sum_{j>r} \sigma_j(T_K)^2 \right)}_{\mathcal{R}_K^\star(r),\ \textit{population rank-}r\ \textit{risk}} + \underbrace{L\frac{\epsilon_n(T_K)}{\Delta_{K,r} - \epsilon_n(T_K)}}_{\textit{excess-risk bound}}. \tag{13}$$

Proof. See Section B.6 for a detailed proof. □

Remark 3.17 (Risk-reduction levers). The bound (13) isolates two controllable levers given fixed targets and context. *Capacity*: the population rank-r floor $\frac{1}{K}\sum_{j>r}\sigma_j(T_K)^2$ vanishes once $r \ge r_\star(K)$, and is bottlenecked by the architectural ceiling $r = d$; the residual tail $\frac{1}{K}\sum_{j>d}\sigma_j(T_K)^2$ persists whenever $d < r_\star(K)$. *Conditioning*: the finite-sample term shrinks as the retained spectrum of T_K becomes more balanced, monitored by $\kappa_K^2 = \lambda_d(H_K)/\lambda_1(H_K)$. The remaining task-predictability floor $\frac{1}{K}\mathbb{E}\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}} Z\|^2$ is irreducible and tracked by $r_\star(K)$ (Definition 3.3).

Thus, given the context and targets, the two controllable levers are capacity and conditioning. Increasing the bottleneck d reduces the population truncation term until $d = r_\star(K)$. Additionally, better eigenvalue-conditioning of whitened end-to-end predictor T_K, reduces the excess risk bound.

3.6 Predictive Isotropy and Regularization

Building upon Proposition 3.16 and Remark 3.17, we now show that the eigenvalue-balancing condition of Proposition 3.13 corresponds precisely to isotropic covariance of the end-to-end prediction on its rank-d range. Furthermore, there exists a gauge-invariant factorization to select a JEPA predictor-encoder pair enforcing isotropic-Gaussian embedding (Section 3.6.1). Recall the stacked context $Z \in \mathbb{R}^{D_x}$ from Section 3.3, the linear context encoder A from Assumption 2.3, and the whitened end-to-end predictor T_K with $H_K := T_K^\top T_K$ from Proposition 3.9.

The object of interest is the end-to-end prediction $P_K := T_K\widetilde{Z}$, i.e., $P_K = B_{\mathrm{OLS}}^{\mathrm{raw}} Z$ written in whitened coordinates. The rank-d JEPA end-to-end prediction WAZ approximates P_K under the embedding bottleneck (8). Whereas LeJEPA's SIGReg regularizes the law of the embedding AZ toward an isotropic Gaussian prior [5], we target the prediction-aware analogue on P_K. We show that the second-moment quantity of this connection is already a sharp spectral condition on H_K.

Lemma 3.18 (Predictive isotropy)**.** *Let $H_K = T_K^\top T_K$. Assuming $\Sigma_{ZZ} \succ 0$, $\mathbb{E}[\widetilde{Z}] = 0$, and $\operatorname{rank}(T_K) = d$, the end-to-end prediction $P_K = T_K\widetilde{Z}$ has isotropic covariance on its rank-d predictive range if and only if*

$$\lambda_1(H_K) = \lambda_2(H_K) = \cdots = \lambda_d(H_K). \tag{14}$$

Proof. See Section B.7 for a detailed proof. □

We now show predictive isotropy (Lemma 3.18) is the unique minimizer of the excess-risk bound.

Theorem 3.19 (Predictive isotropy minimizes the excess-risk bound)**.** *Fix $\tau > 0$ and consider rank-d operators T_K satisfying $\operatorname{tr}(H_K) = \|T_K\|_F^2 = \tau$. Among all such operators, the isotropic spectrum $\lambda_1(H_K) = \cdots = \lambda_d(H_K) = \tau/d$ minimizes the excess-risk bound in (Proposition 3.16):*

Proof. See Section B.8 for a detailed proof. □

The recovery-bound optimum in Theorem 3.19 is an instance of a broader principle, see Section A.2.

3.6.1 Predictive isotropy subsumes encoder isotropy.

Lemma 3.18 admits a distributional reading that places the present subsection one step further down the JEPA pipeline than LeJEPA's SIGReg. Recall that for a measure μ and a map f, the pushforward $f_\#\mu$ denotes the law of $f(X)$ when $X \sim \mu$. SIGReg regularizes the embedding pushforward $\mathcal{L}(AZ) = A_\#\mathcal{L}(Z)$ toward an isotropic Gaussian prior [5]; the prediction-aware analogue is the end-to-end predictive pushforward

$$\mathcal{L}(P_K) = (T_K)_\#\mathcal{L}(\widetilde{Z}). \tag{15}$$

Remark 3.20 (Relation to encoder isotropic regularization). Isotropy of the predictive pushforward $\mathcal{L}(P_K)$ (15) on its rank-d range is exactly the eigenvalue-balancing condition of $H_K = T_K^\top T_K$ of Lemma 3.18, now lifted from covariance to the full law. The relation to SIGReg is sharpened in the regularization paragraph below.

Within Lemma 3.18's condition, the encoder-predictor factorization remains a structural choice. We specify one that bridges the predictive isotropy with the embedding isotropy of canonical JEPA paradigms [5]. Recall the SVD of the whitened end-to-end predictor, $T_K = U_K S_K V_K^\top$, and let $V_K^\top \widetilde{Z} \in \mathbb{R}^d$ be the *SVD coordinate* of the whitened context.

Lemma 3.21 (Gauge invariance of the rank-d optimum)**.** *Let a gauge $G : \mathbb{R}^d \to \mathbb{R}^d$ be any invertible map. Define a encoder-predictor factorization (Λ_G, F_G)*

$$\Lambda_G \coloneqq G \circ V_K^\top \Sigma_{ZZ}^{-1/2}, \qquad F_G \coloneqq U_K S_K \circ G^{-1}.$$

Then $F_G \circ \Lambda_G$ realizes the population rank-d end-to-end prediction P_K for every choice of G.

By Lemma 3.21, predictive isotropy $H_K = \alpha I_d$ is gauge-invariant, so it constrains the end-to-end map T_K without committing to any embedding distribution. The gauge G is therefore free to be chosen for downstream purposes. In particular, it can be selected to realize either of the two canonical encoder-side regularization paradigms in JEPA: covariance isotropy (whitening [16] or VICReg-style [6]) or Gaussian isotropy (LeJEPA [5], SIGReg-style).

Theorem 3.22 (Subsumes covariance isotropy)**.** *For any orthogonal gauge $G \in \mathbb{R}^{d\times d}$, the encoder-predictor factorization (Λ_G, F_G) from Lemma 3.21 induces an isotropic encoder covariance:*

$$\mathrm{Cov}(\Lambda_G(Z)) = I_d.$$

Proof. See Section B.9 for a detailed proof. □

Theorem 3.22 realizes covariance isotropy with a linear gauge. We next show that a nonlinear gauge can realize the strictly stronger property of isotropic-Gaussian embedding under a mild assumption.

Theorem 3.23 (Subsumes Gaussian isotropy)**.** *Assume $V_K^\top \widetilde{Z}$ has absolutely continuous law on $\mathbb{R}^d$. Then there exists an invertible gauge $G^\star : \mathbb{R}^d \to \mathbb{R}^d$ such that the encoder-predictor factorization $(\Lambda_{G^\star}, F_{G^\star})$ from Lemma 3.21 induces an isotropic Gaussian embedding:*

$$\Lambda_{G^\star}(Z) \;\sim\; \mathcal{N}(0, I_d).$$

Proof Sketch. Existence of $G^\star$ follows from Brenier's theorem [17], since the law of $V_K^\top \widetilde{Z}$ admits a density on $\mathbb{R}^d$ by hypothesis and $\mathcal{N}(0, I_d)$ admits a density on $\mathbb{R}^d$. The encoder-predictor pair $(\Lambda_{G^\star}, F_{G^\star})$ then satisfies the conclusion by construction (Lemma 3.21). □

Corollary 3.24 (Linear-Gaussian special case)**.** *If the context follow a Gaussian distribution $Z \sim \mathcal{N}(0, \Sigma_{ZZ})$, the encoder-predictor factorization reduces to the linear pair*

$$\Lambda_I = V_K^\top \Sigma_{ZZ}^{-1/2}, \qquad F_I = U_K S_K.$$

Proof. The whitened context $\widetilde{Z} = \Sigma_{ZZ}^{-1/2} Z \sim \mathcal{N}(0, I_{D_x})$ under Gaussian context, and the rows of $V_K^\top$ are orthonormal, so $\Lambda_I Z = V_K^\top \widetilde{Z} \sim \mathcal{N}(0, I_d)$. Hence $G^\star = I$ realizes Theorem 3.23, and the encoder-predictor factorization specializes to (Λ_I, F_I). □

Subsumption in Theorems 3.22 and 3.23 means the converse fails. As encoder isotropy alone does not imply predictive isotropy, since it is compatible with arbitrary predictor conditioning $\kappa_K^2 \in (0, 1]$. Predictive isotropy is therefore strictly stronger, controlling the predictive spectrum that encoder-only regularization cannot reach. Moreover, by Lemma 3.21, every choice of G realizes the same end-to-end prediction $P_K = T_K \widetilde{Z}$ and hence the same optimality condition as Theorem 3.19, so selecting a gauge in Theorems 3.22 and 3.23 costs nothing against the excess-risk bound.

We now turn this principle into a regularizer. The excess-risk bound in Proposition 3.16 motivates isotropy of the gauge-free predictive object H_K. The subsumption results in Theorems 3.22 and 3.23 further show that, under such regularization, encoder-side isotropy need not be imposed separately, nor must a gauge in Lemma 3.21 be chosen manually.

3.6.2 Maximum-entropy regularization induces isotropic-Gaussian embedding.

The excess-risk bound prescribes only the spectral condition (14) on H_K, equivalently covariance isotropy of P_K on its rank-d predictive range. This places candidate regularizers on a two-axis frontier: *fidelity*, whether $\mathcal{R} = 0$ implies (14), and *commitment*, how much distributional structure $\mathcal{R} = 0$ enforces beyond it. The least-committing faithful choice is therefore the maximum-entropy law among distributions satisfying predictive isotropy. At trace τ, Lemma A.5 shows that this law is uniquely $\mathcal{N}(0, \frac{\tau}{d} I_d)$ on the intrinsic predictive range.

Therefore, we propose a maximum-entropy regularizer for the intrinsic predicted output $\widehat{P}_K^{\text{int}} := \widehat{U}_K^\top \widehat{W} \widehat{A} Z \in \mathbb{R}^d$, where $\widehat{U}_K \in \mathbb{R}^{Kd \times d}$ spans the learned rank-d predictive range. By the Cramér–Wold theorem [18], in its hyperspherical form [5, Lemma 3], the target law $\widehat{P}_K^{\text{int}} \sim \mathcal{N}(0, I_d)$ is equivalent to $u^\top \widehat{P}_K^{\text{int}} \sim \mathcal{N}(0, 1)$ for every $u \in \mathbb{S}^{d-1}$. We approximate this all-direction condition by sampling $|\mathbb{A}|$ random unit vectors $\mathbb{A} \subset \mathbb{S}^{d-1}$, projecting the predictor output onto each direction, and averaging the standard-normal Gaussianity statistics T (Epps-Pulley [19]):

$$\mathcal{R}_{\text{PI}}(\widehat{P}_K^{\text{int}}; \mathbb{A}) := \frac{1}{|\mathbb{A}|} \sum_{u \in \mathbb{A}} T\big(\{u^\top \widehat{p}_n\}_{n=1}^N\big)\,, \tag{16}$$

where $\widehat{p}_n := \widehat{U}_K^\top \widehat{W} \widehat{A} z_n$ is the intrinsic predicted output on the n-th sample.

By Lemma 3.18, the covariance part of this target law $\mathcal{N}(0, I_d)$ recovers the predictive-isotropy condition on the nonzero spectrum of H_K; the Gaussian part is the maximum-entropy completion of that covariance constraint. (16) is the predictive analogue of LeJEPA's SIGReg [5], applied to the predictor output rather than the encoder embedding. Where SIGReg constrains $\mathcal{L}(AZ)$ on the input side, $\mathcal{R}_{\text{PI}}$ constrains $\mathcal{L}(\widehat{P}_K^{\text{int}})$ on the output side, the distributional object whose covariance spectrum is tied to the excess-risk bound through $H_K = T_K^\top T_K$ (Theorem 3.19).

(16) regularizes the H_K toward isotropy to minimize the excess-risk bound in Proposition 3.16. In distributional form, this applies the maximum-entropy principle to the intrinsic predictive signal P_K^{int} without manually choosing a gauge in Lemma 3.21. Together with Theorems 3.22 and 3.23, this shows that isotropic-Gaussian embeddings are encoder-side realizations of the same predictive-isotropy principle. This justifies and unifies the two encoder-side regularization paradigms in canonical JEPA [16, 6, 5] as factorized instances of the predictive-isotropy condition for empirical risk minimization.

4 Experimental Studies

Setup. We test the theory in a frozen-teacher linear-Gaussian setting where the ground truth predictive subspace is known. Each example has latent state $s \sim \mathcal{N}(0, I_{r_\star})$, context $z = Cs + \sigma_z \epsilon_z$, and stacked targets $Y = (y_1^\top, \ldots, y_K^\top)^\top$ with $y_k = A_k s + \sigma_y \epsilon_k$. The context map C is fixed, so the experiments isolate the effect of target structure, bottleneck rank, and finite-sample conditioning. For rank-growth diagnostics we use the stacked covariance Σ_{YZ}; for bottleneck and post-saturation diagnostics we use the whitened raw predictor $T_K = \Sigma_{YZ}^{(K)} \Sigma_{ZZ}^{-1/2}$.

Models. We compare the unconstrained ordinary least-squares predictor with its rank-d reduced-rank regression (RRR) truncation. The RRR rank d plays the role of the JEPA embedding bottleneck. This removes optimization and encoder dynamics, letting each experiment target one mechanism from the theory.

Data. The target stack is constructed to separate complementarity, capacity, and conditioning. We use ten deterministic seeds and report means with standard-error bands. Table 1 summarizes the main diagnostics; exact generator details and appendix-only controls are in Section C.

Metrics. All paper-facing risk panels use the average per-target normalization of $\mathcal{R}_K$ in (10), so risks are comparable when K changes. We report the quantities appearing in the theory: recovered rank, subspace error, heterogeneity λ_{het}, relative conditioning κ_K^2, effective dimension, and spectral-tail ratios. Population values computed from the known operators are shown as dashed curves; finite-sample estimates are shown as solid curves. The appendix defines the effective-rank thresholds, fixed-reference-stack probes, calibrated constants, and Wedin-bound diagnostics.

Experiment	Sweep	Fixed quantities	Main outcome	Figure
K-saturation	K	$D_x = 160,\ d = r_\star = 64$	target-count saturation	Figure 3
Heterogeneity	$\widehat{\lambda}_{\mathrm{het}}$	$D_x = 64,\ K = 48,\ d = 24$	finite-sample recovery	Figure 4
Bottleneck	d	$D_x = 64,\ r_\star = 24,\ K = 48$	spectral-tail floor	Figure 5
Post-saturation	K	$D_x = 160,\ r_\star(K) = d = 64$	relative conditioning	Figure 1
Predictive isotropy	κ_H^2 of H_K	$D_x = 64,\ r_\star = d = K = 24$	fixed-trace isotropy	Figure 6

Table 1: Main synthetic experiment suite. Appendix diagnostics additionally test embedding-versus-predictive isotropy and gauge factorization.

Objective	Test MSE	Test RMSE	Gap	Time
No reg.	0.0923±0.0017	0.304±0.003	0.0854±0.0017	9.03±0.20 s (1.00×)
Enc. reg.	0.0786±0.0013	0.280±0.002	0.0629±0.0014	12.55±0.11 s (1.39×)
Pred. reg.	**0.0600±0.0003**	**0.245±0.001**	**0.0200±0.0007**	12.23±0.04 s (1.35×)
Pred.+Enc.	0.0609±0.0004	0.247±0.001	0.0227±0.0008	15.80±0.13 s (1.75×)

Table 2: Digit regularizer cost–performance. Metrics use right-half pixels normalized to $[0, 1]$; time is per seed.

Baselines. The main baseline is the population counterpart of each finite-sample metric: it is computed directly from the known operators, without sampling noise. For bottleneck risk, unconstrained OLS is the reference model and the predicted floor is the spectral tail beyond the retained rank.

Results. Figures 3 to 5 validate the pre-saturation and bottleneck mechanisms. Complementary targets increase recovered rank and lower per-target MSE until rank saturates. Larger measured $\widehat{\lambda}_{\mathrm{het}}$ improves finite-sample recovery. Sweeping d confirms the capacity prediction: model rank follows $\min\{d, r_\star\}$ and excess risk tracks the spectral-tail floor.

Figure 1 isolates the second, finite-sample saturation mechanism. Here $r_\star(K) = d$ throughout, so additional targets cannot create new rank. They still help when they improve the relative conditioning $\kappa_K^2 = \lambda_d(H_K)/\lambda_1(H_K)$ of the already accessible predictive directions. As κ_K^2 increases, subspace error, excess risk, and finite-sample risk decrease; the half-gap and calibrated-bound checks are reported in Section C.

Figure 6 tests the fixed-trace isotropy prediction. With rank and $\mathrm{tr}(H_K)$ fixed, only the eigenvalue profile of $H_K = T_K^\top T_K$ changes. Consistent with Lemma 3.18 and Theorem 3.19, flatter spectra increase the weakest retained signal relative to perturbation, reduce $\|\sin\Theta_T\|_{\mathrm{op}}$, and lower the balanced-reference-stack excess MSE. Appendix diagnostics show that this predictive isotropy is distinct from embedding isotropy, and that the gauge-factorization claims in Section 3.6.1 hold under both Gaussian and non-Gaussian contexts. Finally, Figure 2 visualizes the real-image digit-half prediction task, and Table 2 reports the corresponding ten-seed quantitative diagnostic. The proposed prediction-aware regularizer reduces low-label target-half pixel MSE and the train–test gap while also improving the prediction-side Gaussianity and covariance-balance diagnostics.

5 Discussion and Conclusion

To be filled.

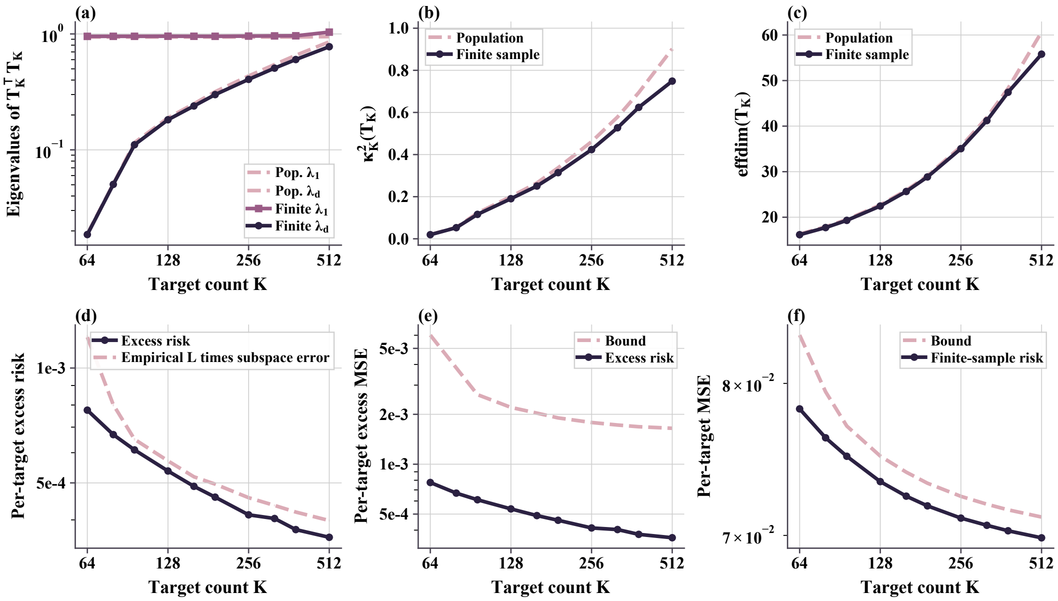

Figure 1: Post-saturation conditioning. With $r_\star(K) = d$ fixed, allocating targets to weak directions increases κ_K^2 and reduces finite-sample risk diagnostics.

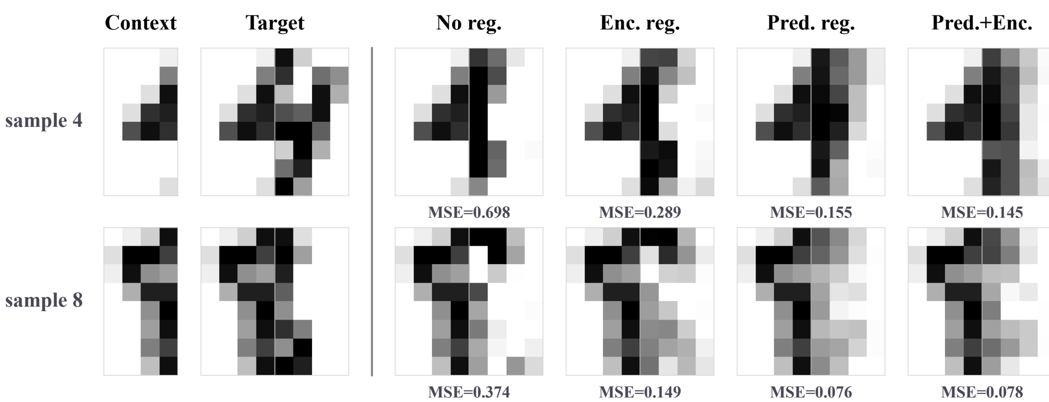

Figure 2: Digit-half completions. Rows are held-out samples; columns show context, ground truth, and candidate completions. Cell labels report per-sample target-half pixel MSE.

Impact Statement

By the theoretical nature of this work, we do not anticipate any negative social impact.

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Appendix

A Supplementary Theoretical Backgrounds

A.1 Reduced-Rank Regression

Given paired random vectors $x \in \mathbb{R}^p$ and $y \in \mathbb{R}^q$, the population squared-loss regression problem

$$\min_{B \in \mathbb{R}^{q\times p}} \mathbb{E}\big[\|y - Bx\|^2\big] \tag{17}$$

admits the closed-form ordinary least squares (OLS) minimizer $B_{\mathrm{OLS}} \coloneqq \Sigma_{YX}\Sigma_{XX}^{-1}$ when $\Sigma_{XX} \succ 0$.

Reduced-rank regression (RRR) [15, 20] restricts the minimization in (17) to $\mathrm{rank}(B) \le r$.

Definition A.1 (Reduced-rank regression problem)**.** Given paired $x \in \mathbb{R}^p$, $y \in \mathbb{R}^q$ as predictor and response random vectors, for a prescribed rank $r \le \min(p, q)$, the reduced-rank regression is

$$B_{\mathrm{RRR}} \in \arg\min_{B \in \mathbb{R}^{q\times p}} \mathbb{E}\big[\|y - Bx\|^2\big] \quad \text{subject to} \quad \mathrm{rank}(B) \le r. \tag{18}$$

The constraint $\mathrm{rank}(B) \le r$ means the linear predictor factors through an r-dimensional latent subspace. The following lemma gives a closed-form solution.

Lemma A.2 (RRR solution, Theorem 2.2 of [20])**.** *Let $V_r \in \mathbb{R}^{p\times r}$ collect eigenvectors corresponding to the top r eigenvalues of $\Sigma_{XX}^{1/2} B_{\mathrm{OLS}}^\top B_{\mathrm{OLS}} \Sigma_{XX}^{1/2}$. A solution to (18) is*

$$B_{\mathrm{RRR}} = B_{\mathrm{OLS}} \Sigma_{XX}^{1/2} V_r V_r^\top \Sigma_{XX}^{-1/2}. \tag{19}$$

Equivalently, RRR solution is obtained by taking the best rank-r approximation of $B_{\mathrm{OLS}}\Sigma_{XX}^{1/2}$ and then unwhitening by $\Sigma_{XX}^{-1/2}$. Informally, this means that RRR retains the r predictor-space directions that are most useful for linearly predicting y.

Remark A.3 (Connection to CCA and PCA). Reduced-rank regression is closely related to canonical correlation analysis (CCA): after appropriate whitening, the rank-r solution retains the leading canonical predictive directions linking the predictor space to the response space. As a special case, when $y = x$, the corresponding reduced-rank self-prediction problem recovers the principal subspace of Σ_{XX}, linking RRR to PCA.

A.2 Maximum-Entropy Laws

We recall two standard entropy facts used in the maximum-entropy regularization argument. Let X be an absolutely continuous random vector in $\mathbb{R}^d$ with density p_X. Its differential entropy is

$$h(X) \coloneqq -\int_{\mathbb{R}^d} p_X(x) \log p_X(x)\, dx,$$

whenever the integral is well-defined.

Lemma A.4 (Gaussian maximum entropy under fixed covariance)**.** *Let X be an absolutely continuous random vector in $\mathbb{R}^d$ with mean μ and covariance $\Sigma \succ 0$. Then*

$$h(X) \le \frac{1}{2}\log\big((2\pi e)^d \det \Sigma\big),$$

with equality if and only if $X \sim \mathcal{N}(\mu, \Sigma)$.

Proof. Let $G \sim \mathcal{N}(\mu, \Sigma)$. Since $D_{\mathrm{KL}}(\mathcal{L}(X) \,\|\, \mathcal{L}(G)) \ge 0$,

$$-h(X) - \mathbb{E}\log p_G(X) \ge 0.$$

Because X and G have the same mean and covariance,

$$-\mathbb{E}\log p_G(X) = \frac{1}{2}\log\big((2\pi)^d \det\Sigma\big) + \frac{1}{2}\mathbb{E}[(X-\mu)^\top\Sigma^{-1}(X-\mu)] = \frac{1}{2}\log\big((2\pi e)^d \det\Sigma\big).$$

Hence the claimed bound follows. Equality holds iff the KL divergence is zero, i.e. iff X has the Gaussian law. □

Lemma A.5 (Trace-constrained maximum entropy)**.** *Let X be an absolutely continuous random vector in $\mathbb{R}^d$ with mean zero, covariance $C \succ 0$, and $\mathrm{tr}(C) = \tau$. Let $h(X)$ denote its differential entropy. Then*

$$h(X) \le \frac{1}{2}\log\big((2\pi e)^d \det C\big) \le \frac{d}{2}\log\left(2\pi e\,\frac{\tau}{d}\right).$$

Equality holds if and only if $X \sim \mathcal{N}(0, \frac{\tau}{d} I_d)$.

Proof. By Lemma A.4, applied with $\Sigma = C$ and $\mu = 0$,

$$h(X) \le \frac{1}{2}\log\big((2\pi e)^d \det C\big),$$

with equality iff $X \sim \mathcal{N}(0, C)$. For the second inequality, arithmetic-geometric mean (AM-GM) inequality gives

$$\det C = \prod_{i=1}^{d} \lambda_i(C) \leq \left(\frac{1}{d} \sum_{i=1}^{d} \lambda_i(C) \right)^d = \left(\frac{\tau}{d} \right)^d,$$

with equality iff $\lambda_1(C) = \cdots = \lambda_d(C) = \tau/d$, i.e., $C = \frac{\tau}{d} I_d$. Combining the equality conditions gives $X \sim \mathcal{N}(0, \frac{\tau}{d} I_d)$. □

Connection to predictive isotropy. Theorem 3.19 is the recovery-bound instance of the same fixed-trace principle: at fixed signal energy $\mathrm{tr}(H_K) = \tau$, the flat spectrum $(\tau/d, \ldots, \tau/d)$ is the unique majorization-minimum among positive rank-d spectra with total mass τ [21, 22]. Equivalently, by AM-GM, it maximizes the log-determinant term $\frac{1}{2} \sum_{i=1}^{d} \log \lambda_i(H_K)$ among fixed-trace spectra. The finite-sample condition $\kappa_K^2 \to 1$ is therefore the empirical manifestation of uniform predictive-energy allocation across the rank-d range.

A.3 Intermediate Derivation of Post-Saturation Recovery

This appendix collects the subspace-recovery results that underpin Proposition 3.13 in the body. The derivation proceeds in two steps. First, a Wedin-type perturbation bound translates the relative conditioning of Assumption 3.11 into a finite-sample $\sin \Theta$ bound on the top-d right singular subspace of T_K (Proposition A.6). Second, a half-gap condition exposes the explicit K-rate hidden inside that bound (Corollary A.7). The body result is then assembled in Proposition 3.13 by composing this $\sin \Theta$ bound with the subspace-Lipschitz Assumption 3.12.

Step 1: Wedin sin-Θ bound.

Proposition A.6 (Post-saturation recovery under relative conditioning)**.** *Under Assumption 3.11, for any $K \in \{K^\star, \ldots, K_{\max}\}$ and any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\| \sin \Theta(V_K, \widehat{V}_K) \|_{\mathrm{op}} \leq \frac{C \sqrt{\frac{r_{\mathrm{eff}} + \log(1/\delta)}{n}}}{\sqrt{\kappa_0^2 + \gamma(K - K^\star)} - C \sqrt{\frac{r_{\mathrm{eff}} + \log(1/\delta)}{n}}}, \tag{20}$$

whenever the denominator is positive.

Proof. By Assumption 3.11, $\mathrm{rank}(T_K) = d$, so the Wedin gap $g_d(T_K)$ reduces to $\sigma_d(T_K)$. By Lemma A.9 and Lemma A.10 applied to T_K and $\widehat{T_K}$,

$$\| \sin \Theta(V_K, \widehat{V}_K) \|_{\mathrm{op}} \leq \frac{\epsilon_n}{\sigma_d(T_K) - \epsilon_n}, \tag{21}$$

where

$$\epsilon_n = C \|T_K\|_{\mathrm{op}} \sqrt{\frac{\mathrm{effdim}(T_K) + \log(1/\delta)}{n}}. \tag{22}$$

Because $\sigma_d(T_K) = \kappa_K \sigma_1(T_K) = \kappa_K \|T_K\|_{\mathrm{op}}$ by (9), we can divide numerator and denominator of (21) by $\sigma_1(T_K) > 0$, obtaining

$$\| \sin \Theta(V_K, \widehat{V}_K) \|_{\mathrm{op}} \leq \frac{C \sqrt{\frac{\mathrm{effdim}(T_K) + \log(1/\delta)}{n}}}{\kappa_K - C \sqrt{\frac{\mathrm{effdim}(T_K) + \log(1/\delta)}{n}}}.$$

Substituting $\mathrm{effdim}(T_K) \leq r_{\mathrm{eff}}$ in the numerator and $\kappa_K \geq \sqrt{\kappa_0^2 + \gamma(K - K^\star)}$ in the denominator both from Assumption 3.11 yields (20). □

Step 2: Explicit K-rate via the half-gap condition. The bound (20) couples the finite-sample term and the deterministic floor in a single ratio, obscuring the K-dependence. To expose the K-rate, we impose a *half-gap condition* requiring the finite-sample term to be at most half the relative singular-value gap:

$$C \sqrt{\frac{r_{\mathrm{eff}} + \log(1/\delta)}{n}} \leq \frac{1}{2} \sqrt{\kappa_0^2 + \gamma(K - K^\star)}. \tag{23}$$

so that the bound simplifies to an explicit rate in K as shown below. This condition is mild in n, and we verify in Section 4 that it is non-vacuous across the K-range studied.

Corollary A.7 (Finite-window post-saturation rate)**.** *Under Assumption 3.11 and the half-gap condition (23), for any $K \in \{K^\star, \ldots, K_{\max}\}$ and any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\|\sin\Theta(V_K, \widehat{V}_K)\|_{\mathrm{op}} \le \frac{2C}{\sqrt{\kappa_0^2 + \gamma(K - K^\star)}}\sqrt{\frac{r_{\mathrm{eff}} + \log(1/\delta)}{n}}. \tag{24}$$

Proof. Substituting the lower bound in (23) into (20) gives (24). □

A.4 Useful Lemmas

Lemma A.8 (Cross-covariance concentration [23])**.** *Let $(X_i, Y_i)_{i=1}^n$ be i.i.d. samples with $X \in \mathbb{R}^d$ and $Y \in \mathbb{R}^{Kd}$, and assume X and Y are sub-Gaussian with bounded second moments. Let $\widehat{\Sigma}_{YX} := \frac{1}{n}\sum_{i=1}^n Y_i X_i^\top$ denote the empirical cross-covariance. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\|\widehat{\Sigma}_{YX} - \Sigma_{YX}\|_{\mathrm{op}} \le C\|\Sigma_{YX}\|_{\mathrm{op}}\sqrt{\frac{\mathrm{effdim}(\Sigma_{YX}) + \log(1/\delta)}{n}},$$

where $C > 0$ is an absolute constant and $\mathrm{effdim}(\Sigma_{YX}) := \frac{\|\Sigma_{YX}\|_F^2}{\|\Sigma_{YX}\|_{\mathrm{op}}^2}$ is the effective rank of Σ_{YX}.

Proof sketch. Apply the matrix Bernstein inequality [23] to the i.i.d. sum $\widehat{\Sigma}_{YX} - \Sigma_{YX} = \frac{1}{n}\sum_{i=1}^n (Y_i X_i^\top - \Sigma_{YX})$ of centered random matrices, with variance proxy controlled by the sub-Gaussian norms of X and Y. Replacing the ambient dimension Kd with the intrinsic effective rank $\mathrm{effdim}(\Sigma_{YX})$ follows from the refinement of [24]. □

Lemma A.9 (Concentration of the whitened raw predictor)**.** *Let $T_K := B_{\mathrm{OLS}}^{\mathrm{raw}}\Sigma_{ZZ}^{1/2} = \Sigma_{YZ}\Sigma_{ZZ}^{-1/2}$ and $\widehat{T}_K := \widehat{\Sigma}_{YZ}\widehat{\Sigma}_{ZZ}^{-1/2}$. Under the same sub-Gaussian assumptions as Lemma A.8, and assuming $\lambda_{\min}(\Sigma_{ZZ}) > 0$, there exists a constant $C > 0$ such that, for any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\|\widehat{T}_K - T_K\|_{\mathrm{op}} \le C\|T_K\|_{\mathrm{op}}\sqrt{\frac{\mathrm{effdim}(T_K) + \log(1/\delta)}{n}}. \tag{25}$$

Proof. Decompose

$$\widehat{T}_K - T_K = (\widehat{\Sigma}_{YZ} - \Sigma_{YZ})\widehat{\Sigma}_{ZZ}^{-1/2} + \Sigma_{YZ}\big(\widehat{\Sigma}_{ZZ}^{-1/2} - \Sigma_{ZZ}^{-1/2}\big).$$

The first term is controlled by the sub-Gaussian cross-covariance concentration bound from Lemma A.8 and the event $\|\widehat{\Sigma}_{ZZ}^{-1/2}\|_{\mathrm{op}} \le C\|\Sigma_{ZZ}^{-1/2}\|_{\mathrm{op}}$. For the second term, standard perturbation bounds for the inverse square root, together with sub-Gaussian concentration of $\widehat{\Sigma}_{ZZ}$, give

$$\|\widehat{\Sigma}_{ZZ}^{-1/2} - \Sigma_{ZZ}^{-1/2}\|_{\mathrm{op}} \le C\|\Sigma_{ZZ}^{-1/2}\|_{\mathrm{op}}\sqrt{\frac{\mathrm{effdim}(T_K) + \log(1/\delta)}{n}}.$$

Combining the two bounds by the triangle inequality and using $T_K = \Sigma_{YZ}\Sigma_{ZZ}^{-1/2}$ yields (25). □

Lemma A.10 (Wedin-type subspace perturbation [25])**.** *Let $M \in \mathbb{R}^{m\times p}$, let $r \le \min(m,p)$, and let $\widehat{M}$ be a perturbation with $\|\widehat{M} - M\|_{\mathrm{op}} \le \epsilon$. Denote by $V_r, \widehat{V}_r$ the top-r right singular subspaces of M and $\widehat{M}$, and define the* Wedin gap at level r *as $g_r(M) := \sigma_r(M) - \sigma_{r+1}(M)$, with the convention $\sigma_{r+1}(M) := 0$ when $\mathrm{rank}(M) \le r$. Provided $\epsilon < g_r(M)$,*

$$\|\sin\Theta(V_r, \widehat{V}_r)\|_{\mathrm{op}} \;\le\; \frac{\epsilon}{g_r(M) - \epsilon}.$$

Proof. Apply Wedin's $\sin\Theta$ theorem [25] to M and $\widehat{M}$ with spectral gap $g_r(M)$. □

B Proofs of Main Text

B.1 Proof of Rank of Reduced-Rank Solution

Proof of Lemma 3.2. Using Lemma A.2 and $B_{\mathrm{OLS}} = \Sigma_{YX}\Sigma_{XX}^{-1}$,

$$W_r^\star = \Sigma_{YX}\Sigma_{XX}^{-1/2}\, V_r V_r^\top\, \Sigma_{XX}^{-1/2}.$$

Since $\Sigma_{XX} \succ 0$, the factor $\Sigma_{XX}^{-1/2}$ is invertible, so

$$\mathrm{rank}(W_r^\star) = \mathrm{rank}\left(\Sigma_{YX}\Sigma_{XX}^{-1/2}\, V_r V_r^\top\right).$$

Let $\Sigma_{YX}\Sigma_{XX}^{-1/2} = U\ \mathrm{diag}(\sigma_1, \ldots, \sigma_{\mathrm{rank}(\Sigma_{YX})}, 0, \ldots, 0)\, V^\top$ be an SVD. By the identity

$$\Sigma_{XX}^{1/2} B_{\mathrm{OLS}}^\top B_{\mathrm{OLS}} \Sigma_{XX}^{1/2} = \big(\Sigma_{YX}\Sigma_{XX}^{-1/2}\big)^\top\big(\Sigma_{YX}\Sigma_{XX}^{-1/2}\big),$$

the columns of V_r are the top-r right singular vectors of $\Sigma_{YX}\Sigma_{XX}^{-1/2}$. Since V is orthogonal, $V_rV_r^\top$ projects onto the top-r right-singular subspace, so

$$\Sigma_{YX}\Sigma_{XX}^{-1/2}\,V_rV_r^\top = U\ \operatorname{diag}(\sigma_1,\ldots,\sigma_{\min(r,\,\operatorname{rank}(\Sigma_{YX}))},0,\ldots,0)\,V^\top,$$

which has rank $\min\{r,\,\operatorname{rank}(\Sigma_{YX})\}$. Using the fact that $\operatorname{rank}(\Sigma_{YX}\Sigma_{XX}^{-1/2}) = \operatorname{rank}(\Sigma_{YX})$ (again because $\Sigma_{XX}^{-1/2}$ is invertible) yields (5). □

B.2 Proof of Rank Growth

Proof of Theorem 3.4. From (6),

$$G_{K+1} = G_K + \Sigma_{Y^{(K+1)}X}^\top\Sigma_{Y^{(K+1)}X}, \tag{26}$$

so for every $v \in \mathbb{R}^d$,

$$v^\top G_{K+1}v = v^\top G_K v + \|\Sigma_{Y^{(K+1)}X}\,v\|^2.$$

Both terms on the right are nonnegative, so $v^\top G_{K+1}v = 0$ holds if and only if both $v^\top G_K v = 0$ and $\Sigma_{Y^{(K+1)}X}v = 0$. Hence

$$\ker(G_{K+1}) = \ker(G_K)\,\cap\,\ker(\Sigma_{Y^{(K+1)}X}) \subseteq \ker(G_K).$$

Therefore

$$\dim\ker(G_{K+1}) \le \dim\ker(G_K).$$

Since $G_K \in \mathbb{R}^{d\times d}$, rank-nullity gives

$$\operatorname{rank}(G_{K+1}) = d - \dim\ker(G_{K+1}) \ge d - \dim\ker(G_K) = \operatorname{rank}(G_K).$$

Using $\operatorname{rank}(G_K) = \operatorname{rank}(\Sigma_{YX}) = r_\star(K)$, we obtain $r_\star(K+1) \ge r_\star(K)$, proving (1).

For (2), suppose $\ker(G_K) \not\subseteq \ker(\Sigma_{Y^{(K+1)}X})$ and pick $v \in \ker(G_K)\setminus\ker(\Sigma_{Y^{(K+1)}X})$. Then $v \in \ker(G_K)$ but $v \notin \ker(G_{K+1})$, so the inclusion above is strict, giving $r_\star(K+1) > r_\star(K)$. □

B.3 Proof of Heterogeneity and Identifiability

Proof of Proposition 3.8. Lemma A.8 gives $\|\widehat{\Sigma}_{YX} - \Sigma_{YX}\|_{\mathrm{op}} \le \epsilon_n$ with probability $1-\delta$. Since $\operatorname{rank}(\Sigma_{YX}) = r_\star(K)$, applying Lemma A.10 with $r = r_\star(K)$, $\epsilon = \epsilon_n$, and Wedin gap $g_{r_\star}(\Sigma_{YX}) = \sqrt{\lambda_{\mathrm{het}}}$ (7) yields the claim. □

B.4 Proof of Capacity Lower Bound

Proof of Proposition 3.9. For any pair (W, A), define the end-to-end map

$$B := WA \in \mathbb{R}^{Kd\times D_x}.$$

Since $W \in \mathbb{R}^{Kd\times d}$ and $A \in \mathbb{R}^{d\times D_x}$, one has $\operatorname{rank}(B) \le d$. Conversely, every matrix $B \in \mathbb{R}^{Kd\times D_x}$ with $\operatorname{rank}(B) \le d$ admits a factorization $B = WA$ for some such (W, A). Hence

$$\inf_{W,A}\mathbb{E}\left[\|Y - WAZ\|_2^2\right] = \inf_{\operatorname{rank}(B)\le d}\mathbb{E}\left[\|Y - BZ\|_2^2\right]. \tag{27}$$

Next, by population least squares, $B_{\mathrm{OLS}}^{\mathrm{raw}} = \Sigma_{YZ}\Sigma_{ZZ}^{-1}$ is the unconstrained minimizer of $\mathbb{E}[\|Y - BZ\|_2^2]$. Therefore, for every $B \in \mathbb{R}^{Kd\times D_x}$,

$$\mathbb{E}\left[\|Y - BZ\|_2^2\right] = \mathbb{E}\left[\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}}Z\|_2^2\right] + \left\|(B - B_{\mathrm{OLS}}^{\mathrm{raw}})\Sigma_{ZZ}^{1/2}\right\|_F^2.$$

Thus

$$\inf_{\operatorname{rank}(B)\le d}\mathbb{E}\left[\|Y - BZ\|_2^2\right] = \mathbb{E}\left[\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}}Z\|_2^2\right] + \inf_{\operatorname{rank}(B)\le d}\left\|B\Sigma_{ZZ}^{1/2} - T_K\right\|_F^2.$$

Since $\Sigma_{ZZ}^{1/2}$ is invertible, the rank constraint remains $\operatorname{rank}(B\Sigma_{ZZ}^{1/2}) \le d$. By the Eckart-Young theorem [26] for Frobenius-norm low-rank approximation,

$$\inf_{\operatorname{rank}(B)\le d}\left\|B\Sigma_{ZZ}^{1/2} - T_K\right\|_F^2 = \sum_{j>d}\sigma_j(T_K)^2. \tag{28}$$

Combining this with (27) yields

$$\inf_{W,A}\mathbb{E}\left[\|Y - WAZ\|_2^2\right] = \mathbb{E}\left[\|Y - B_{\mathrm{OLS}}^{\mathrm{raw}}Z\|_2^2\right] + \sum_{j>d}\sigma_j(T_K)^2. \tag{29}$$

The claimed inequality follows immediately. □

B.5 Proof of Post-Saturation Risk Decomposition

Proof of Proposition 3.13. By definition, $\mathcal{R}_K(\widehat{V}_K) = \mathcal{R}_K^\star + \big(\mathcal{R}_K(\widehat{V}_K) - \mathcal{R}_K^\star\big)$. The first term equals the underbraced architectural floor by Proposition 3.9. For the second, combining Corollary A.7 with the subspace-Lipschitz Assumption 3.12 yields, with probability at least $1-\delta$,

$$\mathcal{R}_K(\widehat{V}_K) - \mathcal{R}_K^\star \le L\,\|\sin\Theta(V_K,\widehat{V}_K)\|_{\mathrm{op}} \le \frac{2LC}{\sqrt{\kappa_0^2 + \gamma(K-K^\star)}}\sqrt{\frac{r_{\mathrm{eff}} + \log(1/\delta)}{n}}. \tag{30}$$

Adding the two bounds yields (12). $\square$

B.6 Proof of Unified Risk Decomposition

Proof of Proposition 3.16. By the reduced-rank regression solution in Lemma A.2, using the whitening argument as in Proposition 3.9, $\mathcal{R}^\star_K(r)$ is exactly the optimal population risk among rank-r end-to-end predictor. Next, on the event $\mathcal{E}$, the Wedin perturbation bound gives

$$\|\sin\Theta(V_{K,r}, \widehat{V}_{K,r})\|_{\mathrm{op}} \le \frac{\epsilon_n(T_K)}{\Delta_{K,r} - \epsilon_n(T_K)},$$

because $\Delta_{K,r} = \sigma_r(T_K) - \sigma_{r+1}(T_K)$ and $\epsilon_n(T_K) < \Delta_{K,r}$. This is the same perturbation step used in Proposition A.6, now applied at rank r. By Assumption 3.12,

$$\mathcal{R}_K(\widehat{V}_{K,r}) - \mathcal{R}^\star_K(r) \le L\|\sin\Theta(V_{K,r}, \widehat{V}_{K,r})\|_{\mathrm{op}}.$$

Substituting the Wedin bound into this inequality gives (13). $\square$

B.7 Proof of Predictive Isotropy

Proof. Let $T_K = U_K S_K V_K^\top$ be a compact singular value decomposition, where $U_K \in \mathbb{R}^{Kd\times d}$ and $V_K \in \mathbb{R}^{D_x\times d}$ have orthonormal columns, and $S_K = \mathrm{diag}(\sigma_1(T_K), \ldots, \sigma_d(T_K))$. Since $\mathrm{rank}(T_K) = d$, $P_K = T_K\widetilde{Z}$ lies in $\mathrm{col}(T_K) = \mathrm{col}(U_K)$. Thus $U_K^\top P_K$ is the intrinsic coordinate vector of P_K on its rank-d predictive range. Covariance isotropy of P_K on this range is therefore equivalent to covariance isotropy of $U_K^\top P_K$ on $\mathbb{R}^d$. Using $P_K = T_K\widetilde{Z}$ and $\mathrm{Cov}(\widetilde{Z}) = I_{D_x}$,

$$\mathrm{Cov}(U_K^\top P_K) = \mathrm{Cov}(U_K^\top T_K\widetilde{Z}) = \mathrm{Cov}(S_K V_K^\top \widetilde{Z}) = S_K^2.$$

Thus P_K has isotropic covariance on its rank-d range if and only if $S_K^2 = cI_d$ for some $c > 0$. The nonzero eigenvalues of $H_K = T_K^\top T_K$ are the diagonal entries of S_K^2, so this is exactly (14). $\square$

B.8 Proof of Predictive Isotropy Minimize the Excess-Risk Bound

Proof of Theorem 3.19. Let $\lambda_1 \ge \cdots \ge \lambda_d > 0$ denote the nonzero eigenvalues of H_K, with $\sum_{i=1}^d \lambda_i = \tau$. At fixed $\mathrm{tr}(H_K) = \tau$, $\lambda_d \le \tau/d \le \lambda_1$, with both equalities if and only if the spectrum is flat; hence the isotropic spectrum simultaneously minimizes λ_1 and maximizes $\lambda_d = \sigma_d(T_K)^2$, which uniquely maximizes $\kappa_K^2 = \lambda_d/\lambda_1 \le 1$. Since $\|T_K\|_{\mathrm{op}}^2 = \lambda_1$ and $\mathrm{effdim}(T_K) = \sum_{i=1}^d \lambda_i/\lambda_1 = \tau/\lambda_1$, and the perturbation scale in Proposition A.6 has the form

$$\epsilon_n(T_K) = C\|T_K\|_{\mathrm{op}}\sqrt{\frac{\mathrm{effdim}(T_K) + \log(1/\delta)}{n}},$$

we obtain

$$\epsilon_n(T_K) = C\sqrt{\frac{\tau + \lambda_1\log(1/\delta)}{n}}.$$

Combining $\epsilon_n \downarrow$ and $\sigma_d \uparrow$, the isotropic spectrum minimizes the Wedin upper bound $\epsilon_n/(\sigma_d - \epsilon_n)$ in (20) whenever the denominator is positive. $\square$

B.9 Proof of Covariance Isotropy Subsumption

Proof of Theorem 3.22. By Lemma 3.21, $\Lambda_G(Z) = GV_K^\top \Sigma_{ZZ}^{-1/2} Z$, so

$$\mathrm{Cov}(\Lambda_G(Z)) = GV_K^\top \Sigma_{ZZ}^{-1/2}\,\Sigma_{ZZ}\,\Sigma_{ZZ}^{-1/2} V_K G^\top = GV_K^\top V_K G^\top = GG^\top = I_d,$$

using $V_K^\top V_K = I_d$ from the SVD and $GG^\top = I_d$ by orthogonality property. $\square$

C Additional Experimental Details

Synthetic generator. Each synthetic instance is generated from a latent linear–Gaussian model. We sample a latent vector $s \in \mathbb{R}^{r_\star}$ and observe

$$z = Cs + \sigma_z\epsilon_z, \qquad y_k = a_k^\top s + \sigma_y\epsilon_k,$$

where $C \in \mathbb{R}^{D_x\times r_\star}$ is a randomly sampled orthonormal context operator, a_k is the kth target direction, and the noise terms are independent isotropic Gaussians. Thus the population predictive structure is known, but the context coordinates are not aligned with the canonical basis.

Target stacks. The target operators are constructed in latent space. In the saturation and bottleneck experiments, we use a complementary target stack whose directions progressively cover the $r_\star$ latent predictive coordinates. In the heterogeneity experiment, the construction parameter is chosen so that the induced population heterogeneity level λ_{het} is sampled approximately evenly; the plot therefore uses the measured heterogeneity level directly on the horizontal axis. In the predictive-isotropy ablation, the target stack is constructed to control the spectrum directly. For a spectrum decay parameter $\rho \in (0, 1]$, we set

$$A^\top A = Q\,\mathrm{diag}(\lambda_1, \ldots, \lambda_d)Q^\top, \qquad \lambda_i = \tau\frac{\rho^{i-1}}{\sum_{j=1}^d \rho^{j-1}},$$

where Q is a random orthogonal rotation and τ is fixed across all conditions. Thus $\operatorname{rank}(A) = d$ and $\operatorname{tr}(A^\top A) = \tau$ are unchanged, while the measured spectrum condition $\kappa_H^2 = \lambda_d(H_K)/\lambda_1(H_K)$ ranges from 1 to approximately 10^{-8}. Since the context operator has orthonormal columns and σ_z is fixed, this fixes the nonzero spectrum of $H_K = T_K^\top T_K$ up to the same scalar context-noise factor across all conditions. The decay parameter ρ is therefore only a generator knob; the figure and interpretation use the measured operator quantities κ_H^2 and entropy gap.

Models and estimators. We first estimate the unrestricted linear predictor by ordinary least squares and then compute its rank-d reduced-rank regression approximation by truncating the singular spectrum. The learned BM-JEPA linear predictor is therefore the best rank-constrained linear map induced by the finite training sample. Population curves use the corresponding population cross-covariance operator under the same target stack and bottleneck dimension.

Metrics. All paper-facing MSE panels report average per-target test mean-squared error, matching the normalization of $\mathcal{R}_K$ in (10). The CSV files also retain fixed-reference-stack diagnostics on a common rich evaluation target stack as auxiliary representation probes, but these are not the main risk quantities used in the theory-facing panels. In Figure 4, we show both quantities because they answer different questions: per-target $\mathcal{R}_K$ evaluates the changing target stack used for training, whereas reference-stack MSE evaluates the recovered subspace on a common rich target stack. As heterogeneity increases, $\mathcal{R}_K$ can remain nearly constant or rise slightly because the task becomes less redundant; the reference-stack probe can decrease at the same time because the recovered subspace becomes more complete. The standard-error band of $\mathcal{R}_K$ also shrinks as target coverage spreads across more latent directions, so the per-target average becomes less dominated by a single shared direction. In Figure 6, panel (d) uses a related reference-stack probe: after estimating the top-d right singular subspace of the whitened operator $\widehat{T}_K$, we fit a linear predictor from that recovered subspace to a fixed flat evaluation target stack. This reference-stack excess MSE is not a new training objective; it is a representation diagnostic showing whether the recovered predictive subspace captures all latent directions equally well. For the embedding-versus-predictive isotropy diagnostic, we additionally assign a synthetic embedding covariance spectrum with eigenvalues proportional to ρ_{emb}^{i-1} and fixed trace. This embedding spectrum is varied independently of the predictive spectrum of H_K. It should therefore be read as a controlled proxy for embedding-level isotropy diagnostics, not as a second training objective. Recovered rank is the effective numerical rank of the learned predictor, computed using a relative singular-value threshold of 0.05 times the leading singular value. The heterogeneity metric λ_{het} is the $r_\star$th nonzero eigenvalue of the target-induced Gram operator. In the latent coordinates this is the same nonzero spectrum as $A^\top A$; in the observed context coordinates it is the same nonzero spectrum as $\Sigma_{ZY}\Sigma_{YZ}$ because $C^\top C = I$. Subspace error is measured by the largest principal-angle sine between the recovered and true predictive subspaces. For the heterogeneity recovery bound, we also compute the empirical perturbation $\widehat{\epsilon} = \|\widehat{\Sigma}_{YZ} - \Sigma_{YZ}\|_{\text{op}}$ and the corresponding Davis–Kahan/Wedin-style denominator $\sqrt{\lambda_{\text{het}}} - \widehat{\epsilon}$. When this denominator is nonpositive, the finite-sample bound is vacuous. We keep these quantities in the CSV as validity diagnostics, but omit them from Figure 4 because the bound is loose for this sweep and is less informative than the rank and risk panels. The sharper recovery-bound visualization appears in the post-saturation experiment, where the relative conditioning condition is non-vacuous over the main finite window.

For the bottleneck experiment, the plotted spectral-tail quantity is the reduced-rank approximation error predicted by the bottleneck spectral-tail result:

$$\mathcal{E}_{\text{tail}}(d) = \sum_{j>d} \sigma_j^2,$$

where $\{\sigma_j\}$ are singular values of the relevant whitened regression operator. In the figure we report the normalized ratio $\mathcal{E}_{\text{tail}}(d)/\sum_j \sigma_j^2$ and compare it with the correspondingly normalized observed excess risk of the rank-d estimator over the unrestricted OLS estimator.

The rank panel in the bottleneck figure reports the effective rank of the learned reduced-rank predictor, not the rank of the stacked cross-covariance. The latter is fixed by the target stack and therefore does not change when the bottleneck dimension is swept. Since the synthetic target stack has latent predictive rank $r_\star = 24$, a well-estimated rank-d predictor is expected to use approximately $\min\{d, r_\star\}$ directions. This panel is therefore a sanity check that the bottleneck cap is the active rank constraint before $d = r_\star$ and that additional capacity is unused afterward.

Table 3: Phase-1 synthetic experiment configurations.

Experiment	n_{train}	n_{test}	D_x	$r_\star$	K	d	Noise
K saturation	1024	8192	160	64	sweep	64	0.12
Heterogeneity	32768	8192	64	24	48	24	0.10
Bottleneck	128	8192	64	24	48	sweep	0.16
Post-saturation	16384	8192	160	64	sweep	64	0.25
Predictive isotropy	1024	8192	64	24	24	24	0.25
Embedding vs. predictive isotropy	1024	8192	64	24	24	24	0.25
Gauge factorization	8192	8192	64	24	24	24	0.25

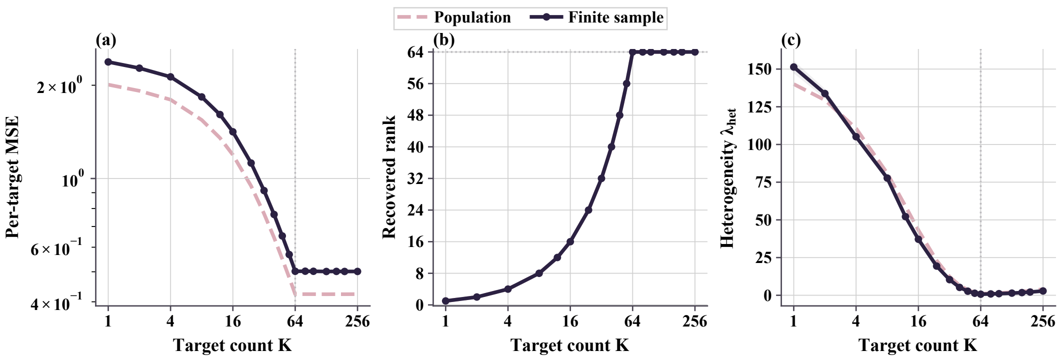

Figure 3: K-saturation diagnostic. Complementary targets reduce per-target MSE and increase recovered rank until predictive rank saturates.

Baselines. The main finite-sample baseline is unrestricted OLS, which removes the embedding bottleneck. Population curves are included as reference predictions for the same synthetic distribution. The bottleneck experiment additionally reports a plug-in spectral tail computed from the finite-sample OLS spectrum, allowing the risk decomposition to be checked using learned empirical quantities rather than only population quantities.

Configuration summary. We list the main Phase-1 configurations used for the reported figures in Table 3.

Additional theory diagnostics. The CSV schema also records the effective rank of the learned RRR coefficient, the finite-sample target value $\min\{d, \widehat{r}_\star(K)\}$ predicted by the rank identity, and their difference. This directly checks the rank identity beyond the displayed rank of the stacked cross-covariance.

The diagnostic in Figure 1 tests the post-saturation relative-conditioning mechanism. The target stack is full rank from $K^\star = d = 64$ onward, so additional targets do not change $r_\star(K)$. The relevant post-saturation conditioning ratio is

$$\kappa_K^2 = \frac{\lambda_d(H_K)}{\lambda_1(H_K)}, \qquad H_K = T_K^\top T_K, \quad T_K = \Sigma_{YZ}^{(K)} \Sigma_{ZZ}^{-1/2}.$$

The initial saturated target stack is deliberately imbalanced. Writing $A_K^\top A_K$ for the latent target-stack design Gram, we initialize

$$A_{K^\star}^\top A_{K^\star} = Q \operatorname{diag}(c_1, \dots, c_{64}) Q^\top,$$

where Q is a fixed random orthogonal rotation and

$$c_i = 0.02^{(i-1)/63}.$$

Thus the stack is already rank-64, but $\kappa_{K^\star}^2 \approx 0.02$. For $K > K^\star$, each additional target row is added to the currently weakest eigendirection and contributes 0.1 units of target energy. This keeps $r_\star(K) = d$ throughout the plotted window while increasing κ_K^2. The finite-sample subspace error and per-target excess MSE decrease over the same window, supporting the repaired post-saturation claim that extra targets help only when they improve relative conditioning of the already-accessible predictive subspace. In Figure 1, we keep the core post-saturation diagnostics in the first row: the weakest and top eigenvalues of $H_K = T_K^\top T_K$ in a shared panel, the relative conditioning ratio κ_K^2, and the effective dimension of T_K. The second row reports the risk-stability bound, excess risk, and

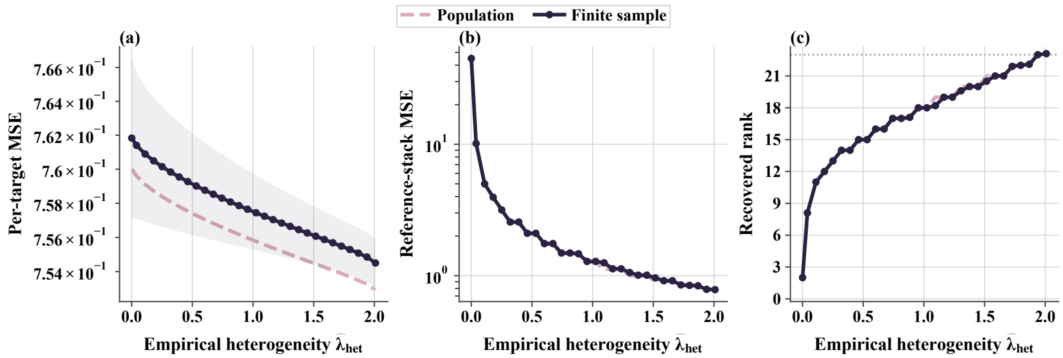

Figure 4: Target-heterogeneity diagnostic. Larger $\widehat{\lambda}_{\text{het}}$ improves rank recovery; the reference-stack probe measures transfer to a common target stack.

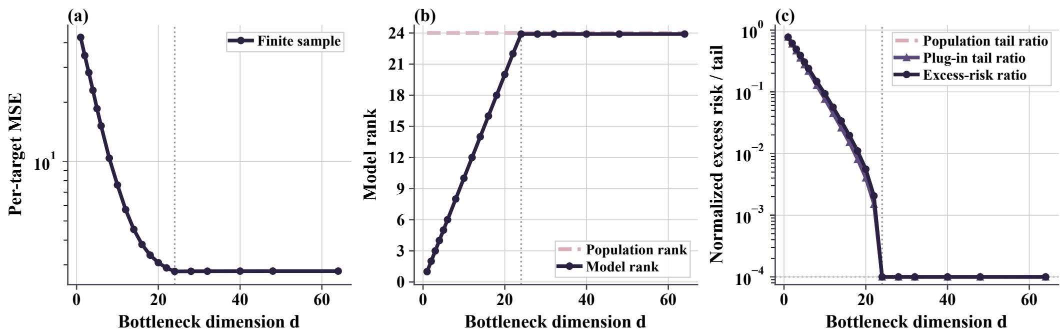

Figure 5: Embedding-bottleneck diagnostic. As d increases, model rank follows $\min\{d, r_\star\}$ and excess risk tracks the spectral-tail ratio.

finite-sample risk. The effective-dimension panel checks that $\operatorname{effdim}(T_K)$ remains bounded over the finite window in the finite-window post-saturation relative-geometry condition. To instantiate the risk-stability constant, we compute the empirical ratio

$$\widehat{L}_K = \frac{[\widehat{\mathcal{R}}_K - \mathcal{R}_K^\star]_+}{\|\sin\Theta(V_K, \widehat{V}_K)\|_{\text{op}}},$$

using per-target excess MSE in the numerator, and set $\widehat{L} = \max_{K,\text{seed}} \widehat{L}_K$. Panel (d) then plots the excess-risk left-hand side against the empirical bound curve $\widehat{L}\|\sin\Theta(V_K, \widehat{V}_K)\|_{\text{op}}$, matching the form of the normalized risk-stability condition. Each seed is one deterministic draw of the synthetic data and target stack, and the same $\widehat{L}$ is used across the full plotted window. In the current run this gives $\widehat{L} = 7.07 \times 10^{-3}$.

For the concentration constant in (22), we set $\delta = 0.1$ and compute

$$a_K = \sqrt{\frac{\operatorname{effdim}(T_K) + \log(1/\delta)}{n}}, \qquad \widehat{\eta}_K = \frac{\|\widehat{T}_K - T_K\|_{\text{op}}}{\|T_K\|_{\text{op}}}.$$

We then set $\widehat{C}_{0.9}$ to the 90th percentile, over all plotted (K, seed) pairs, of $\widehat{\eta}_K / a_K$. This gives $\widehat{C}_{0.9} = 1.79$ and covers $90/100$ normalized perturbation samples by construction. Panel (e) uses the calibrated bound

$$\frac{2\widehat{L}\widehat{C}_{0.9}a_K}{\kappa_K}$$

from (30). Panel (f) adds this term to the population OLS risk estimate, matching the risk-decomposition bound in Proposition 3.13. In this post-saturation setting $d = r_\star(K)$, so the spectral-tail term is zero up to numerical precision.

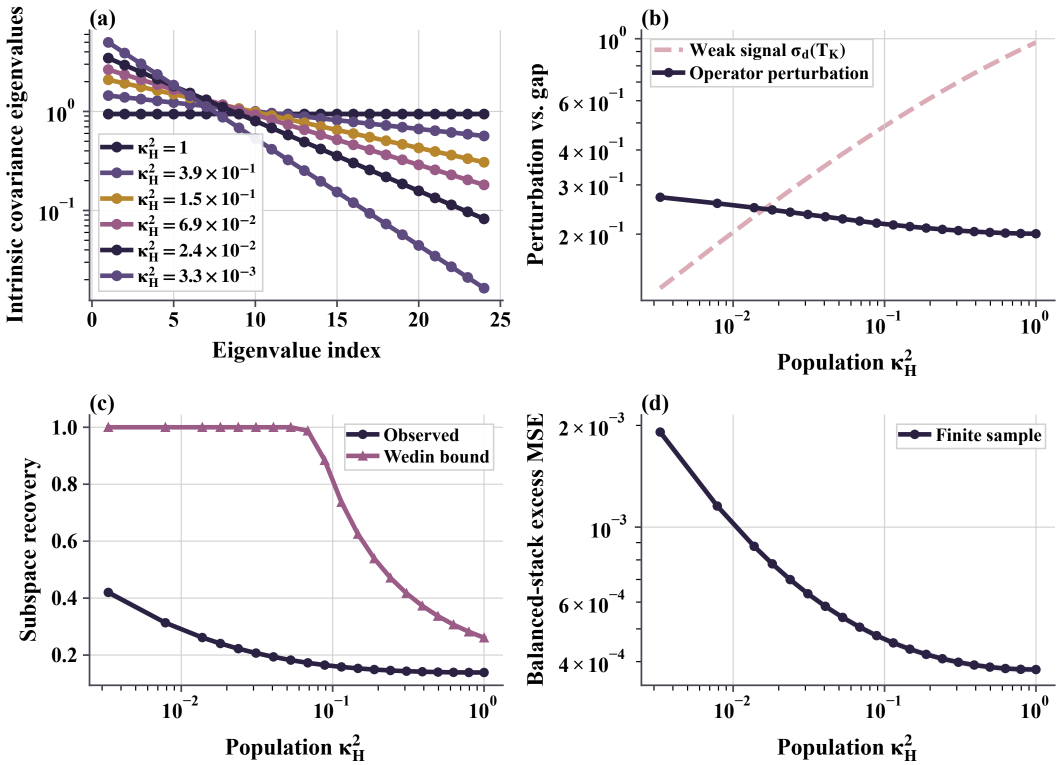

Figure 6: Predictive-isotropy diagnostic. At fixed rank and trace, flatter spectra of $H_K = T_K^\top T_K$ improve recovery and reduce balanced-stack excess MSE.

To check that the simplified rate is not vacuous in the plotted regime, we evaluate the empirical analogue of the half-gap condition in (23). Define the normalized perturbation

$$\widehat{\eta}_K := \frac{\|\widehat{T}_K - T_K\|_{\mathrm{op}}}{\|T_K\|_{\mathrm{op}}}.$$

The condition $\widehat{\eta}_K \le \kappa_K/2$ holds seed-wise for all deterministic runs in the plotted range. The weakest endpoint, $K = 64$, is deliberately closest to the small-conditioning boundary and remains just inside the half-gap window for the current sample size. Representative values are:

K	$\widehat{\eta}_{K,\mathrm{mean}}$	$\widehat{\eta}_{K,\max}$	$\kappa_K/2$	valid seeds
64	0.0613	0.0656	0.0707	10/10
96	0.0626	0.0688	0.1751	10/10
512	0.0929	0.0947	0.4748	10/10

Predictive-isotropy construction. The fixed-trace predictive-isotropy experiment is designed to test the second-moment prediction in Lemma 3.18 and the fixed-trace bound minimization in Theorem 3.19, without changing rank or total signal energy. All conditions use $D_x = 64$, $r_\star = d = K = 24$, $\sigma_z = \sigma_y = 0.25$, and $\mathrm{tr}(H_K) = 22.59$ after the common whitening/context-noise scaling. The generator uses 20 decay values between $\rho = 1.00$ and $\rho = 0.78$, concentrated near the transition where the weakest retained signal becomes comparable to the empirical perturbation scale. They induce a dense sweep of population conditioning values $\kappa_H^2 \in [3.30\times10^{-3}, 1]$. Representative points are:

ρ	κ_H^2	entropy gap	$\|\sin\Theta_T\|_{\mathrm{op}}$
1.00	1.0000	0.000	0.139
0.94	2.41×10^{-1}	−0.090	0.146
0.90	8.86×10^{-2}	−0.253	0.165
0.86	3.12×10^{-2}	−0.495	0.207
0.84	1.81×10^{-2}	−0.644	0.241
0.78	3.30×10^{-3}	−1.191	0.420

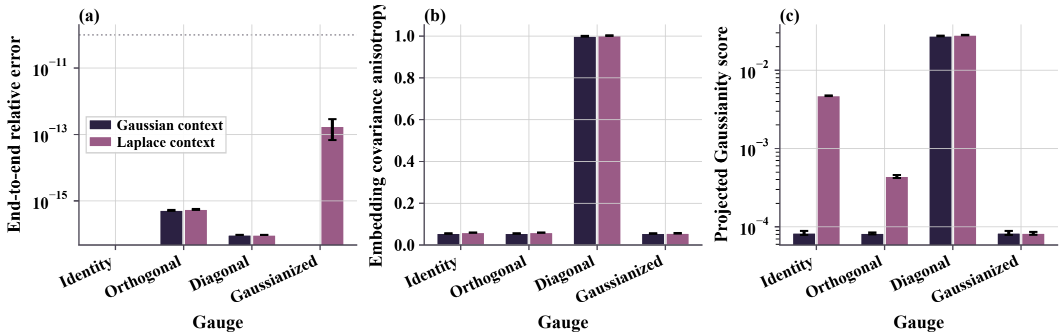

Figure 7: Gauge-factorization diagnostic. Gauge changes preserve prediction while changing encoder covariance and Gaussianity diagnostics.

The trace column is omitted because it is constant by construction. Since $\widetilde{Z} = \Sigma_{ZZ}^{-1/2} Z$ is Gaussian in this generator, the intrinsic covariance of $P_K = T_K \widetilde{Z}$ on its rank-d range has eigenvalues $\lambda_i(H_K)$. Thus panel (a) of Figure 6 is directly the covariance spectrum appearing in Lemma 3.18. Panel (b) plots the two quantities that determine the perturbative denominator: $\sigma_d(T_K)$ and $\|\widehat{T}_K - T_K\|_{\mathrm{op}}$. Panel (c) compares the observed subspace recovery error with the Wedin bound $\|\widehat{T}_K - T_K\|_{\mathrm{op}} / (\sigma_d(T_K) - \|\widehat{T}_K - T_K\|_{\mathrm{op}})$, clipped at one when the denominator is nonpositive. The small-κ_H^2 settings intentionally enter this vacuous-bound regime; they show the transition from perturbative recovery to the boundary where weak directions begin to become difficult to recover. In the current ten-seed run, 16 of the 20 spectrum settings have a positive perturbative denominator for every seed, and 10 of 20 satisfy the half-gap diagnostic for every seed.

Gauge-factorization diagnostic. The gauge diagnostic directly targets Section 3.6.1. For each seed, we construct the population whitened predictor $T_K = U_K S_K V_K^\top$ and compare identity, random orthogonal, anisotropic diagonal, and nonlinear Gaussianizing gauges under both Gaussian and standardized Laplace SVD-coordinate distributions. For an invertible linear gauge G, the synthetic encoder and predictor are

$$\Lambda_G(Z) = G V_K^\top \Sigma_{ZZ}^{-1/2} Z, \qquad F_G(\cdot) = U_K S_K G^{-1}(\cdot).$$

Panel (a) of Figure 7 verifies Lemma 3.21: all gauges reproduce the same end-to-end prediction up to numerical precision. Panel (b) checks the covariance part of Theorem 3.22: identity and orthogonal gauges remain covariance-isotropic up to finite-sample error for both context laws, while the diagonal gauge is deliberately anisotropic. Panel (c) reports a Cramer–Wold-style projected standard-normal discrepancy. For Gaussian context, the identity/SVD-aligned gauge already realizes the linear-Gaussian special case in Corollary 3.24. For Laplace context, identity remains covariance-isotropic but not Gaussian; the nonlinear Gaussianizing gauge maps the intrinsic coordinates to standard-normal marginals and sharply reduces the projected discrepancy, illustrating the role of the nonlinear gauge in Theorem 3.23.

Regularization implication. The revised theory proposes a prediction-aware analogue of embedding-side Gaussian regularization. Standard embedding regularizers act on the marginal pushforward $\mathcal{L}(AZ)$, whereas the theory-facing object here is the intrinsic end-to-end predictive pushforward from Lemma 3.18. For a minibatch, project the learned predicted outputs onto the learned rank-d predictive range and collect the intrinsic coordinates into rows $\widehat{p}_n^{\mathrm{int}}$ of

$$\widehat{P}^{\mathrm{int}} \in \mathbb{R}^{N \times d}.$$

The proposed regularizer samples random directions $\mathcal{A} \subset \mathbb{S}^{d-1}$ and averages a univariate Gaussianity test, such as the Epps–Pulley statistic, over the projected samples:

$$\mathcal{R}_{\mathrm{PI}}(\widehat{P}^{\mathrm{int}}; \mathcal{A}) = \frac{1}{|\mathcal{A}|} \sum_{u \in \mathcal{A}} \mathcal{T}\big(\{u^\top \widehat{p}_n^{\mathrm{int}}\}_{n=1}^N\big)\,.$$

This is a Cramer–Wold-style, prediction-aware counterpart to SIGReg: it acts on intrinsic predictor outputs rather than encoder embeddings. Matching projected standard normals is stronger than the second-moment condition $H_K = \alpha I$, but it implies the isotropic predictive covariance targeted by

Lemma 3.18. A practical nonlinear objective could therefore take the schematic form

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{emb}}\mathcal{L}_{\text{emb-iso}} + \lambda_{\text{pred}}\mathcal{R}_{\text{PI}}.$$

The first regularizer is task-agnostic and prevents marginal representation collapse; the second is task-conditioned and targets the predictive spectrum appearing in the finite-sample recovery and excess-risk bounds. The gauge results in the theory explain why a sufficiently strong prediction-aware regularizer can subsume encoder-side isotropy at the population level; in practice, the two terms may still be combined for optimization and collapse prevention.

The experiment in Figure 6 validates the second-moment spectral mechanism behind this proposal. The next diagnostic tests the corresponding regularizer in a learned nonlinear predictor.

Digits regularizer experiment. We add a low-label real-data training diagnostic on the `sklearn` handwritten-digits dataset. The context is the left half of the 8×8 digit image and the target is the right half. A nonlinear encoder maps the context to a 16-dimensional embedding; a predictor maps this embedding to a 16-dimensional intrinsic prediction representation; a final linear head predicts the target pixels. We identify this predictor representation with the finite-network analogue of $\widehat{P}_K^{\text{int}}$ in (16); the pixel head is only the readout used to evaluate target prediction. To make generalization rather than in-sample reconstruction the limiting factor, we train a high-capacity model on only an 8% stratified training split and evaluate on the remaining examples. We compare four objectives: no regularizer, an encoder-only Gaussianity penalty, a prediction-only Gaussianity penalty, and a prediction-plus-encoder auxiliary condition. The projected-Gaussianity term combines the Cramer–Wold/Epps–Pulley statistic with a covariance-to-identity penalty, and is applied either to the encoder embedding or to the prediction representation. The prediction-plus-encoder condition is included as a control: the theory predicts that the prediction-side condition is the risk-relevant one, so adding an encoder-side auxiliary penalty should not be necessary for the main risk effect.

Figure 8 and Table 2 show the result over ten deterministic seeds. Here MSE is the primary criterion. It is the mean squared error over right-half target pixels after normalizing digit intensities to $[0, 1]$; it is not a percentage. The corresponding RMSE can be read as a fraction of the full gray-scale range. The Gaussianity score is only a diagnostic: it is a distance from projected standard-normal laws, so smaller values mean the regularizer has moved the intended distribution, but this score is not a substitute for test risk. Effective rank is likewise a conditioning diagnostic, not a standalone objective. In this harder setting, prediction-side regularization reduces test MSE from $9.23{\times}10^{-2}$ to $6.00{\times}10^{-2}$ and the train–test gap from $8.54{\times}10^{-2}$ to $2.00{\times}10^{-2}$. The same runs reduce predictive Gaussianity distance from $1.14{\times}10^{-1}$ to $4.33{\times}10^{-2}$ and raise predictive effective rank from 4.13 to 6.41. Encoder-side regularization also improves test MSE, but less strongly, and it does not improve the prediction-side Gaussianity distance. The prediction-only and prediction-plus-encoder conditions have nearly identical test MSE, which is the expected control outcome under the subsumption result. The timing panel makes the cost tradeoff explicit: prediction-side regularization takes 12.23 seconds per seed on average, compared with 9.03 seconds for no regularizer and 15.80 seconds for the combined objective. Thus the prediction-side term gives essentially the same test risk as the combined condition at lower cost. This experiment is not meant to replace large-scale JEPA validation, but it does test the proposed regularizer inside an actual learned nonlinear predictor rather than a closed-form synthetic operator. Figure 2 gives a qualitative view of the same task: the left half of each test digit is observed, and each method predicts the right half.

Reproducibility. All Phase-1 experiments are run with ten deterministic seeds. The full synthetic suite can be reproduced with

```
python -m experiments.run_all
```

followed by

```
python -m plots.plot_main.
```

The qualitative digit reconstruction figure is generated separately with

```
python -m plots.plot_regularizer_digits_samples.
```

The scripts write per-experiment CSV files, the combined result table, and the main and appendix PDF figures. Unit tests cover deterministic generation, OLS/RRR behavior, rank constraints, subspace metrics, CSV schema, and figure-reference existence.

Visualization choices. Finite-sample estimates are plotted as solid curves with standard-error bands. Population predictions are plotted as dashed gray curves. For the bottleneck spectral-tail-ratio panel,

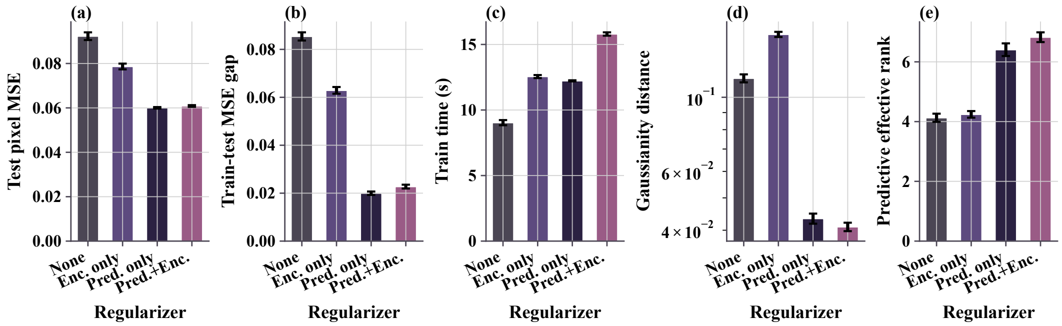

Figure 8: Digit regularizer diagnostics. Prediction-side regularization gives the best pixel-MSE/generalization tradeoff and improves prediction-side conditioning diagnostics.

values below 10^{-4} are visually floored in the log-scale display to keep the super-threshold regime legible; this affects only the display, not the CSV values or computations.

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